

Skull Osteology Practical — Study Guide

This guide combines six sources: the 2024 answer key, the graded 2025 test, the All-Star frequency list (practicals from 1994–2012), the labeled diagram study sheets, the real-skull photo Quizlet (58 cards), and the 251-card diagram Quizlet. Bones follow the 2024 key unless noted.

How to use it: learn Tier 1 cold, then Tier 2 and Tier 3. Use the Atlas (A1–A13) and the real-skull Photos (P1–P58) referenced in each row to see the structure. Then test yourself in the Recall Bank, which gives a word bank to pick from on every page, with answer keys at the end.

All-Star count = how many of the 12 archived practicals (1994–2012) included the structure.

1. Priority list

Tier 1: on both the 2024 and 2025 practicals (33)

Occipital

Structure	Bone(s)	All-Star	Atlas	Photos
Posterior condylar canal	Occipital	2	A4	—
Hypoglossal (anterior condylar) canal	Occipital	11	A5	P3, P4
Inferior nuchal line	Occipital	1	A2	—
Jugular tubercle	Occipital	2	A5	—
Pharyngeal tubercle	Occipital	2	A4	—
Clivus	Occipital	1	A3	—

Temporal

Structure	Bone(s)	All-Star	Atlas	Photos
Tympanic canaliculus	Temporal	1	A6	—
Petrotympenic fissure	Temporal	10	A6	P11
Styloid process	Temporal	11	A5	P5
Digastric notch	Temporal	9	A3, A5	P17
Articular eminence	Temporal	4	A6	—
Carotid canal	Temporal	10	A6	P7
Stylomastoid foramen	Temporal	8	A6	P28

Sphenoid

Structure	Bone(s)	All-Star	Atlas	Photos
Superior orbital fissure	Sphenoid	2	A1, A7	—
Pterygoid canal	Sphenoid	10	A7	P13
Anterior clinoid process	Sphenoid	8	A7	P23
Dorsum sellae	Sphenoid	5	A7	P53

Mandible

Structure	Bone(s)	All-Star	Atlas	Photos
Mental spines	Mandible	10	A12	P9
Mandibular lingula	Mandible	10	A11	P6
External oblique line	Mandible	6	A11	P43
Mandibular notch	Mandible	2	A11	—
Sublingual fossa	Mandible	5	A11	P44

Maxilla

Structure	Bone(s)	All-Star	Atlas	Photos
Incisive canal / foramen	Maxilla	3–5	A3, A9	P49
Anterior nasal spine	Maxilla	5	A1, A9	P46
Posterior superior alveolar foramen	Maxilla	6	A3, A9	P34
Infraorbital foramen	Maxilla	6	A1, A9	P39

Multi-bone and other

Structure	Bone(s)	All-Star	Atlas	Photos
Anterior ethmoidal foramen	Frontal, ethmoid	6	A1, A8	P36
Bregma	Frontal, parietal	11	A2	—
Nasolacrimal duct / canal	Lacrimal, maxilla	4	A1, A12	P57
Greater palatine foramen	Maxilla, palatine	9	A3, A10	P22
Posterior lacrimal crest	Lacrimal	1	A1, A12	—
Crista galli	Ethmoid	1	A8	—
Zygomaticofacial foramen	Zygomatic	4	A1, A11	—

Tier 2: probably on both tests (4)

These were on the 2024 key and likely match 2025 answers marked wrong. This is inferred from the red marks, not confirmed.

Structure	Bone(s)	All-Star	Atlas	Photos
Pterygomaxillary fissure	Sphenoid, maxilla	3	A3	—
Infratemporal crest	Sphenoid, temporal	1	A3, A5	—
Foramen spinosum	Sphenoid	4	A6, A7	P18, P19
Sphenoidal spine	Sphenoid	6	A7	P35

Tier 3: All-Star heavy hitters not on the 2024 or 2025 tests (23)

Each appeared on at least 6 of the 12 archived practicals, so they are the most likely new additions.

Structure	Bone(s)	All-Star	Atlas	Photos
Scaphoid fossa	Sphenoid	11	A7	P1
Mylohyoid groove	Mandible	11	A11	P2
Lambda	Occipital, parietal	10	A2	P8
Mylohyoid line	Mandible	10	A11	P10
Pterygoid fovea	Mandible	10	A11	P12
Foramen ovale	Sphenoid	10	A6, A7	P14, P15
Groove for middle meningeal artery	Parietal or temporal	10	A4	P16
Foramen rotundum	Sphenoid	9	A7	—
Pterygoid hamulus	Sphenoid	9	A7, A8	P20
Groove for sigmoid sinus	Temporal	9	A5	P21
Groove for greater petrosal nerve	Temporal	8	A5	P24
Lateral pterygoid plate	Sphenoid	8	A7	P26
Mental foramen	Mandible	8	A11	P27
Internal occipital crest	Occipital	7	A5	P29
Digastric fossa	Mandible	7	A12	P30
Foramen cecum	Frontal	7	A4	P31
Groove for auditory tube	Temporal	7	—	P32
Prechiasmatic groove	Sphenoid	6	A7	P33
Frontal crest	Frontal	6	A4	P37
Inferior temporal line	Parietal, temporal	6	A2	P38
Trigeminal depression	Temporal	6	A6	P40
Coronoid process	Mandible	6	A11	P41
Palatovaginal canal	Palatine, sphenoid	6	A8	P42

Where the sources disagree

- Clivus: the 2024 key lists occipital only; the All-Star list gives sphenoid/occipital. The 2025 grader took a point off for sphenoid, so write occipital.
- Greater palatine foramen: the 2024 key lists maxilla and palatine; the All-Star list and photo Quizlet list palatine only. Writing both matches the 2024 key.
- Photo Quizlet card 12 says "pterygoid fossa, mandible"; the mandibular structure is the pterygoid fovea (the pterygoid fossa is on the sphenoid). It is corrected in this guide.
- Photo Quizlet card 50 says "inferior occipital fissure"; it is the inferior orbital fissure. Corrected here.
- The All-Star list has two lingulas: mandible (10) and sphenoid (2). Know both.

2. All-Star frequency list (1994–2012)

★ = on both the 2024 and 2025 tests. ☆ = probably on both (Tier 2). Sorted by how many practicals included it.

Structure	Bone(s)	Count	2024/25
Bregma	Parietal/frontal	11	★
Scaphoid fossa	Sphenoid	11	
Mylohyoid groove	Mandible	11	
Hypoglossal canal	Occipital	11	★
Styloid process	Temporal	11	★
Lingula	Mandible	10	★
Carotid canal	Temporal	10	★
Lambda	Occipital/parietal	10	
Mental spine	Mandible	10	★
Mylohyoid line (ridge)	Mandible	10	
Petrotympenic fissure	Temporal	10	★
Pterygoid fovea/fossa	Mandible	10	
Pterygoid canal	Sphenoid	10	★
Foramen ovale	Sphenoid	10	
Groove for middle meningeal artery	Parietal or temporal	10	
Digastric notch/groove (mastoid notch)	Temporal	9	★
Foramen rotundum	Sphenoid	9	
Pterygoid hamulus	Sphenoid	9	
Groove for sigmoid sinus	Temporal	9	
Greater palatine foramen	Palatine	9	★
Anterior clinoid process	Sphenoid	8	★
Groove for greater petrosal nerve	Temporal	8	
Lateral pterygoid plate	Sphenoid	8	
Mental foramen	Mandible	8	
Stylomastoid foramen	Temporal	8	★
Internal occipital crest	Occipital	7	
Digastric fossa	Mandible	7	
Foramen cecum	Frontal	7	
Gutter/groove for auditory tube	Temporal	7	
Prechiasmatic groove	Sphenoid	6	
Posterior superior alveolar foramen	Maxilla	6	★
Sphenoid spine	Sphenoid	6	☆
Anterior ethmoid foramen	Ethmoid/frontal	6	★
Frontal crest	Frontal	6	
Inferior temporal line	Parietal/temporal	6	
Infraorbital foramen	Maxilla	6	★
Trigeminal depression	Temporal	6	
Coronoid process	Mandible	6	
Palatovaginal canal	Palatine/sphenoid	6	

Structure	Bone(s)	Count	2024/25
External oblique line	Mandible	6	★
Groove for the occipital artery	Temporal	5	
Sublingual fossa	Mandible	5	★
Lesser palatine foramen	Palatine	5	
Temporal crest (internal oblique ridge)	Mandible	5	
Wormian bone	Varies	5	
Anterior nasal spine	Maxilla	5	★
Fovea for arachnoid granulation	Parietal/frontal	5	
Groove for superior petrosal sinus	Temporal	5	
Incisive foramen (fossa)	Maxilla	5	★
Inferior orbital fissure	Sphenoid/maxilla	5	
Infraorbital groove	Maxilla	5	
Internal acoustic meatus	Temporal	5	
Dorsum sellae	Sphenoid	5	★
Groove for inferior petrosal sinus	Temporal	5	
Metopic suture	Frontal	5	
Articular eminence	Temporal	4	★
Frontal sinus	Frontal	4	
Foramen spinosum	Sphenoid	4	☆
Mandibular canal	Mandible	4	
Zygomaticotemporal foramen	Zygomatic	4	
Nasolacrimal canal/lacrimal fossa	Lacrimal/maxilla	4	★
Postglenoid tubercle	Temporal	4	
Zygomaticofacial foramen	Zygomatic	4	★
Canine fossa	Maxilla	3	
Hypophyseal (pituitary) fossa	Sphenoid	3	
Incisive canal	Maxilla	3	★
Inion (external occipital protuberance)	Occipital	3	
Mastoid notch	Temporal	3	
Posterior nasal spine	Palatine	3	
Submandibular fossa	Mandible	3	
Supraorbital foramen	Frontal	3	
Tympanic plate	Temporal	3	
Zygomatico-orbital foramen	Zygomatic	3	
Pterygomaxillary fissure	Sphenoid/maxilla	3	☆
Confluens of sinuses	Occipital	3	
Groove for superior sagittal sinus	Occipital/parietal	3	
Hamulus	Sphenoid	3	
Mandibular foramen	Mandible	3	
Pterion	Frontal/parietal/sphenoid/temporal	3	
Spine (sphenoid)	Sphenoid	3	☆

Structure	Bone(s)	Count	2024/25
Infraorbital canal	Maxilla	3	
Lambdoid suture	Occipital/parietal	3	
Anterior lacrimal crest	Maxilla	2	
Articular tubercle	Temporal	2	
External occipital protuberance	Occipital	2	
Carotid groove	Sphenoid	2	
Groove for greater petrosal nerve (facial canal)	Temporal	2	
Lesser palatine canal	Palatine	2	
Mandibular notch	Mandible	2	★
Nasion	Frontal/nasal	2	
Optic canal	Sphenoid	2	
Pterygoid fossa	Sphenoid	2	
Retromolar triangle	Mandible	2	
Superior nuchal line	Occipital	2	
Vomerovaginal canal	Vomer/sphenoid	2	
Coronal suture	Frontal/parietal	2	
Pharyngeal tubercle	Occipital	2	★
Posterior clinoid process	Sphenoid	2	
Squamotympanic fissure	Temporal	2	
External occipital crest	Occipital	2	
Inferior temporal line	Parietal	2	
Jugular tubercle	Occipital	2	★
Lingula (sphenoid)	Sphenoid	2	
Posterior condylar canal	Occipital	2	★
Superior orbital fissure	Sphenoid	2	★
Ala	Vomer	1	
Angle	Mandible	1	
Asterion	Occipital/temporal/parietal	1	
Clivus	Sphenoid/occipital	1	★
Crest	Frontal	1	
Crista galli	Ethmoid	1	★
Depression for confluence of sinuses	Occipital	1	
Depression for the lacrimal gland	Frontal	1	
Depression for the trigeminal ganglion	Temporal	1	
Foramina for PSA nerve and vessels	Maxilla	1	
Greater petrosal groove	Temporal	1	
Greater wing	Sphenoid	1	
Groove for transverse sinus	Occipital	1	
Horizontal/transverse palatine suture	Palatine/maxilla	1	
Incisive fossa	Mandible	1	

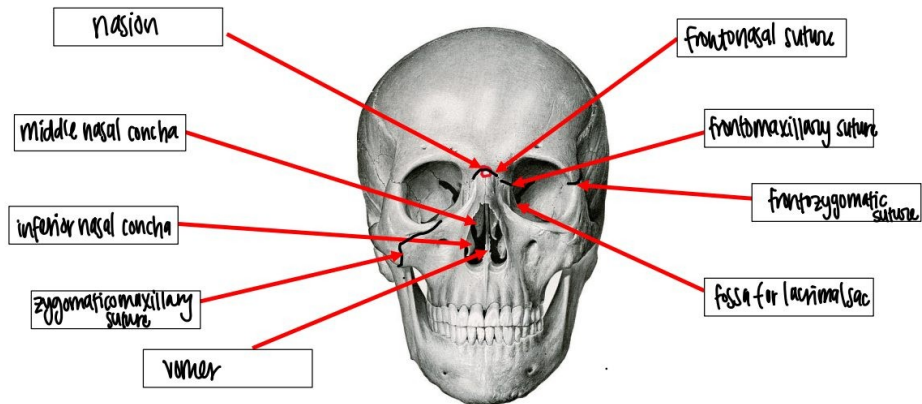
Structure	Bone(s)	Count	2024/25
Inferior nasal concha	Inferior nasal concha	1	
Inferior petrosal sulcus	Temporal	1	
Infratemporal crest	Sphenoid	1	☆
Intermaxillary suture	Maxilla	1	
Lacrimal bone	Lacrimal	1	
Masseteric tuberosity	Mandible	1	
Mastoid air cells	Temporal	1	
Medial pterygoid plate	Sphenoid	1	
Orbital plate	Frontal	1	
Parietal foramina	Parietal	1	
Posterior ethmoidal foramen	Ethmoid	1	
Posterior lacrimal crest	Lacrimal	1	★
Prechiasmatic groove	Sphenoid	1	
Retromolar fossa	Mandible	1	
Sphenoid bone	Sphenoid	1	
Sphenoid sinus	Sphenoid	1	
Stylomastoid canal	Temporal	1	
Superior petrosal sulcus	Temporal	1	
Superior temporal line	Parietal	1	
Supraorbital notch	Frontal	1	
Temporal process of zygomatic bone	Zygomatic	1	
Tegmen tympani	Temporal	1	
Trochlea	Frontal	1	
Tuberculum sellae	Sphenoid	1	
Trochlear spine	Frontal	1	
Tympanic canal	Temporal	1	★
Hiatus for the lesser petrosal nerve	Temporal	1	
Inferior nuchal line	Occipital	1	★
Superior sagittal sulcus	Parietal	1	

3. Labeled atlas (A1–A13)

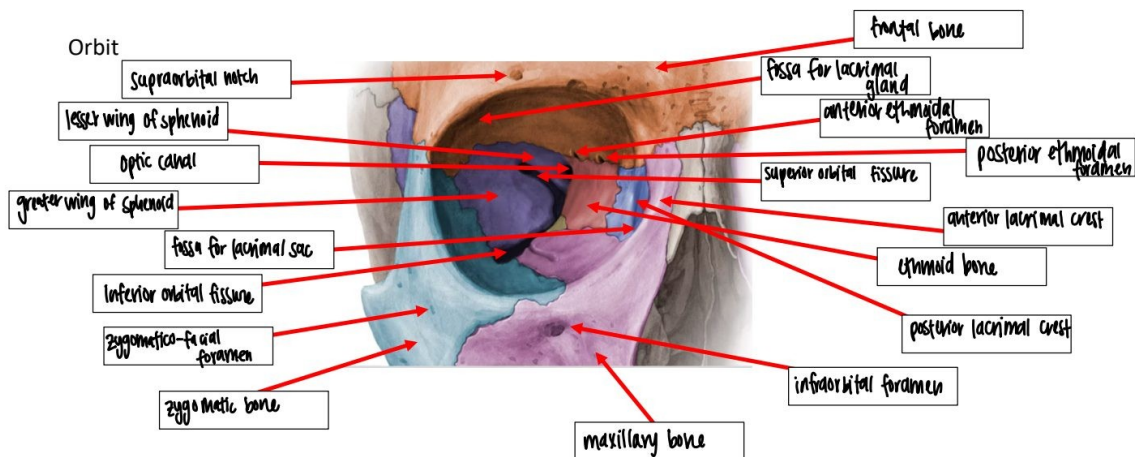
Labeled diagrams from the study sheets. The Recall Bank has the same diagrams with the labels hidden.

A1. Skull frontal view, orbit, nose

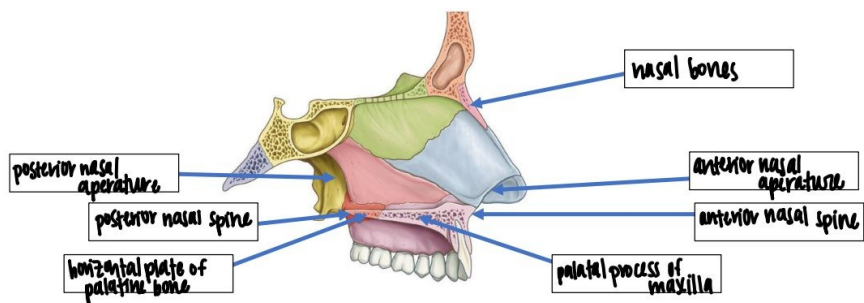
Skull Frontal View



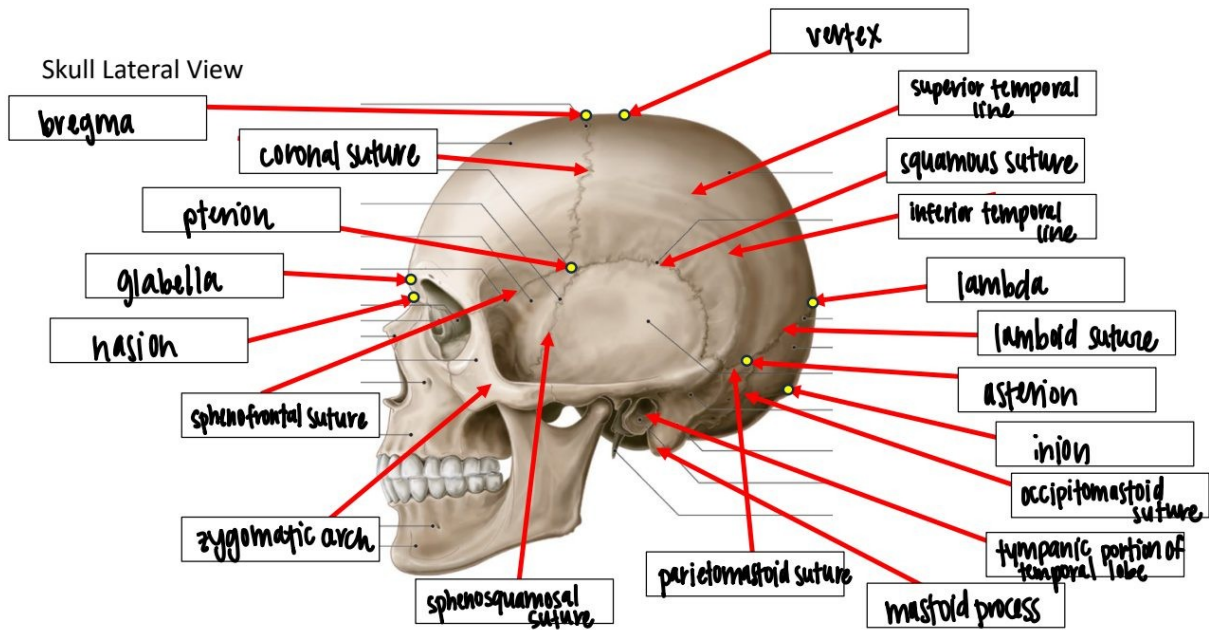
Orbit



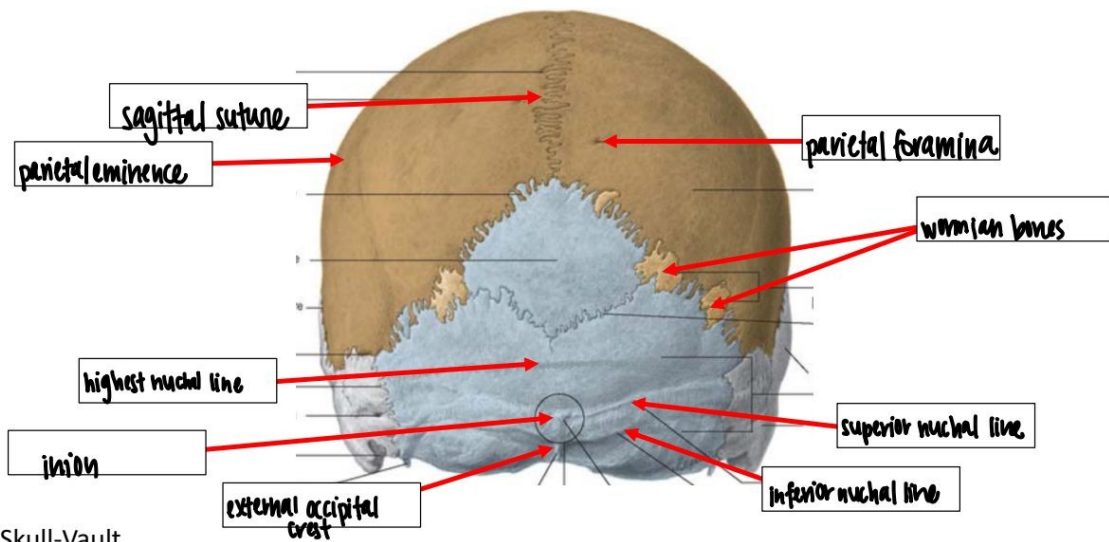
Nose



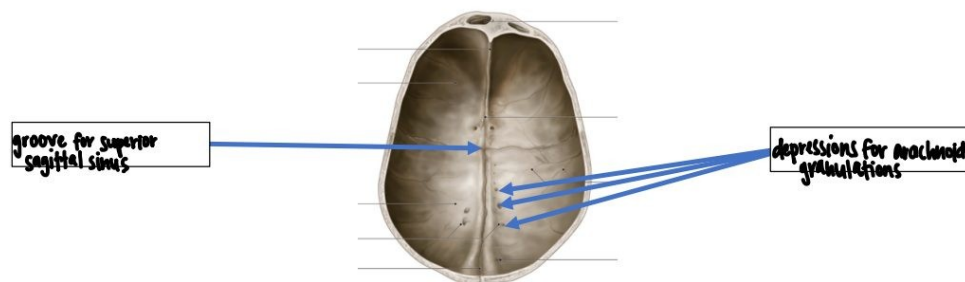
A2. Lateral view, posterior view, vault



Skull Posterior View

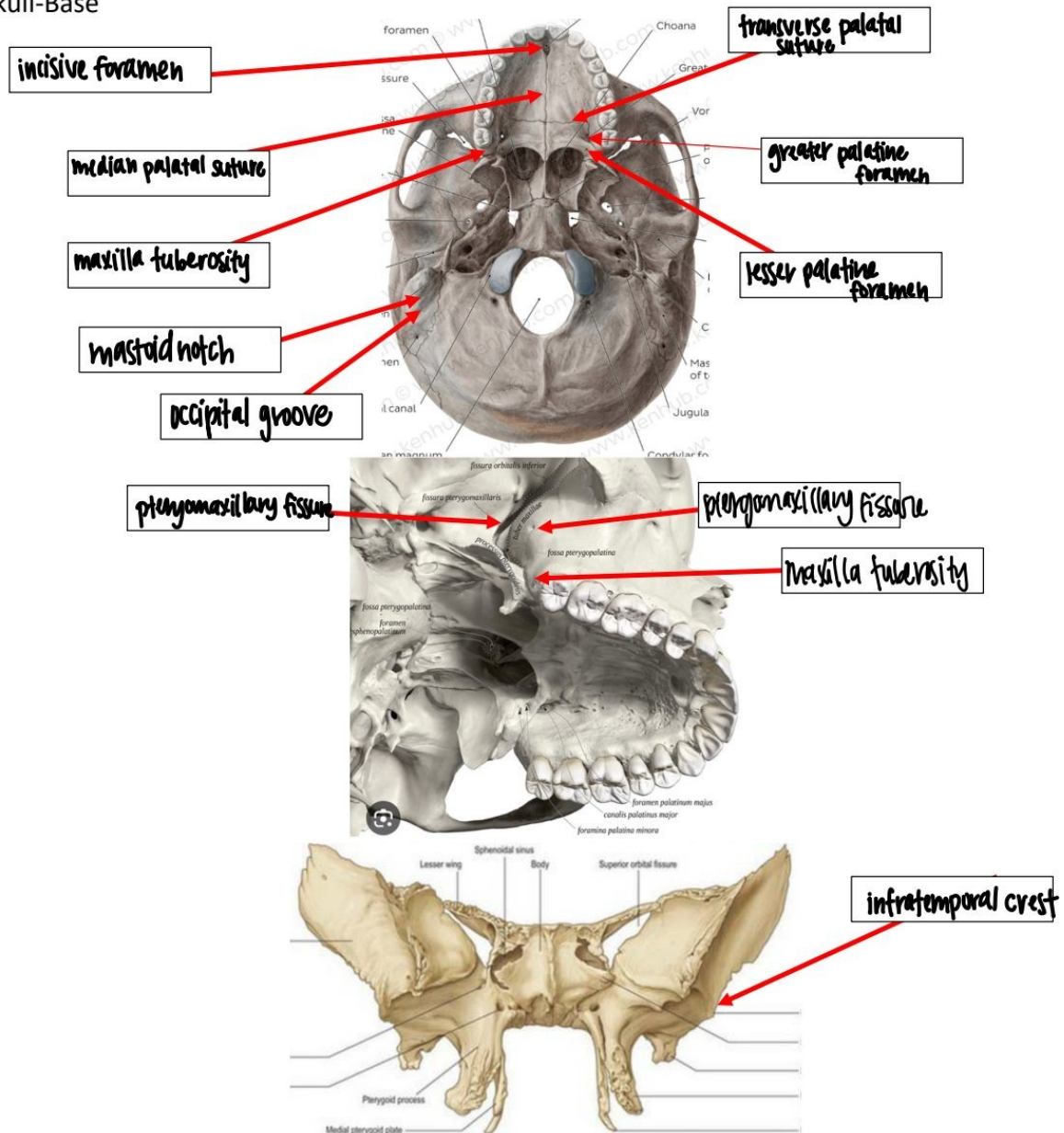


Skull-Vault

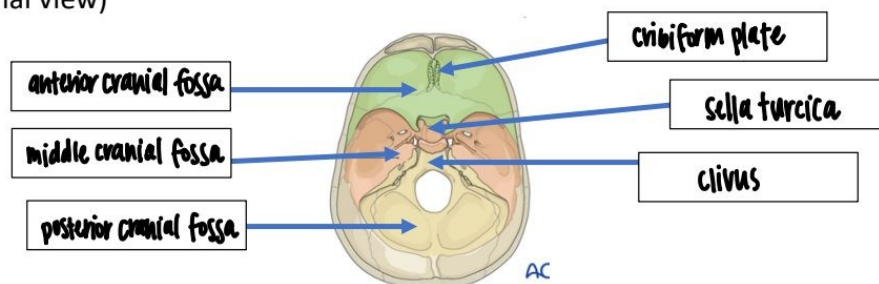


A3. Skull base and cranial fossae

Skull-Base

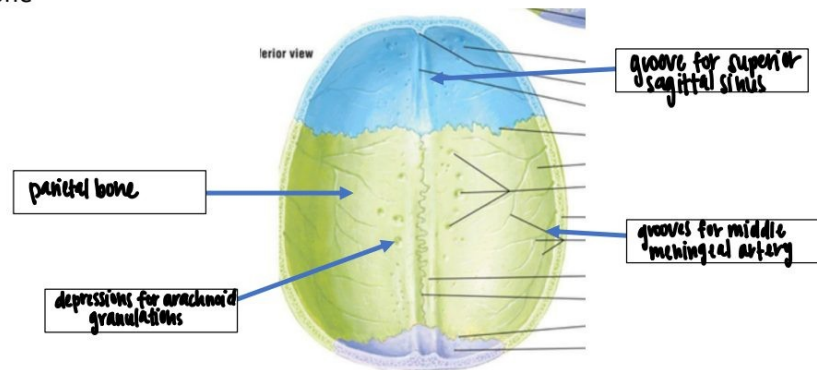


Regions (internal view)

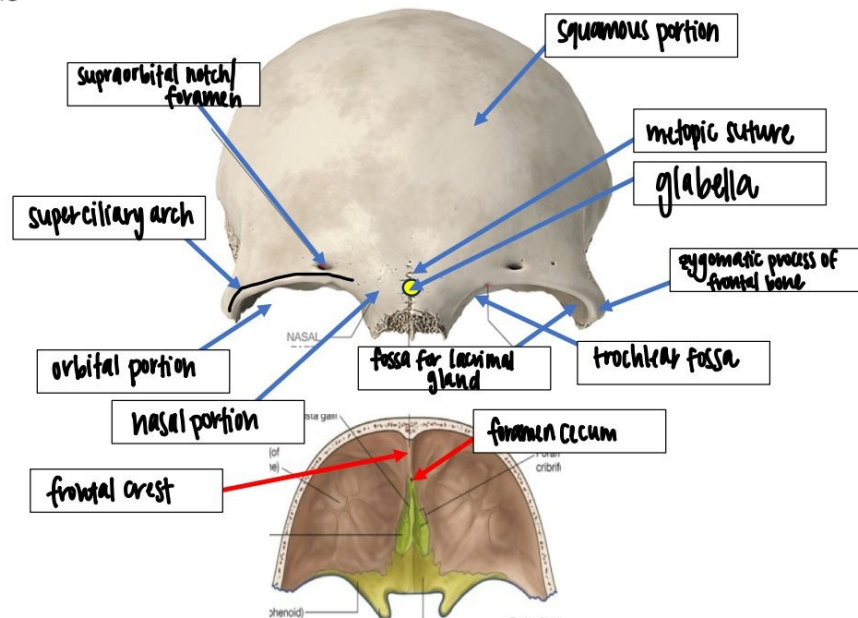


A4. Parietal, frontal, occipital (external)

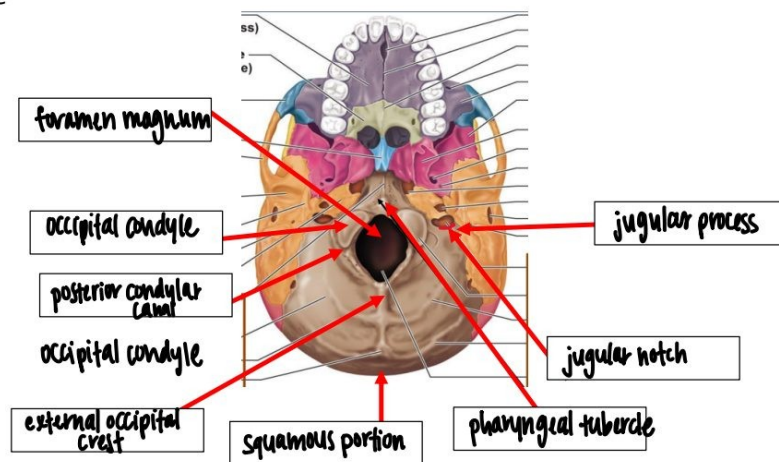
Parietal Bone



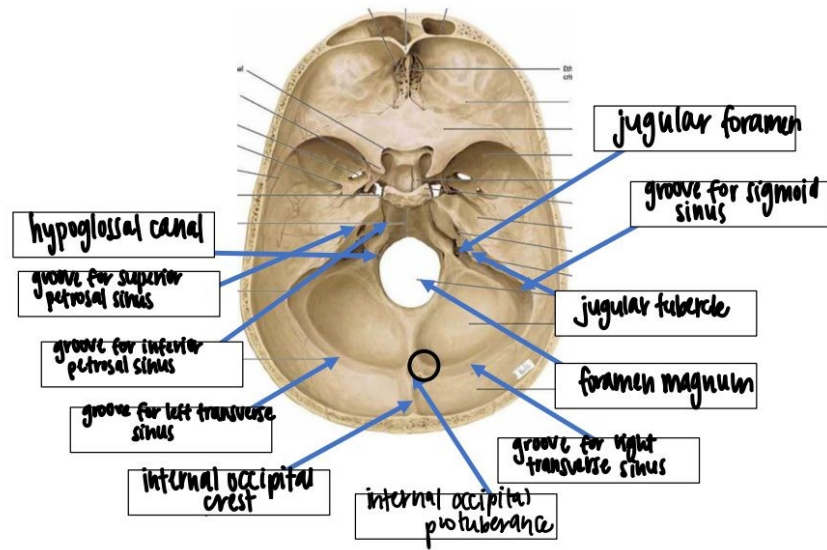
Frontal Bone



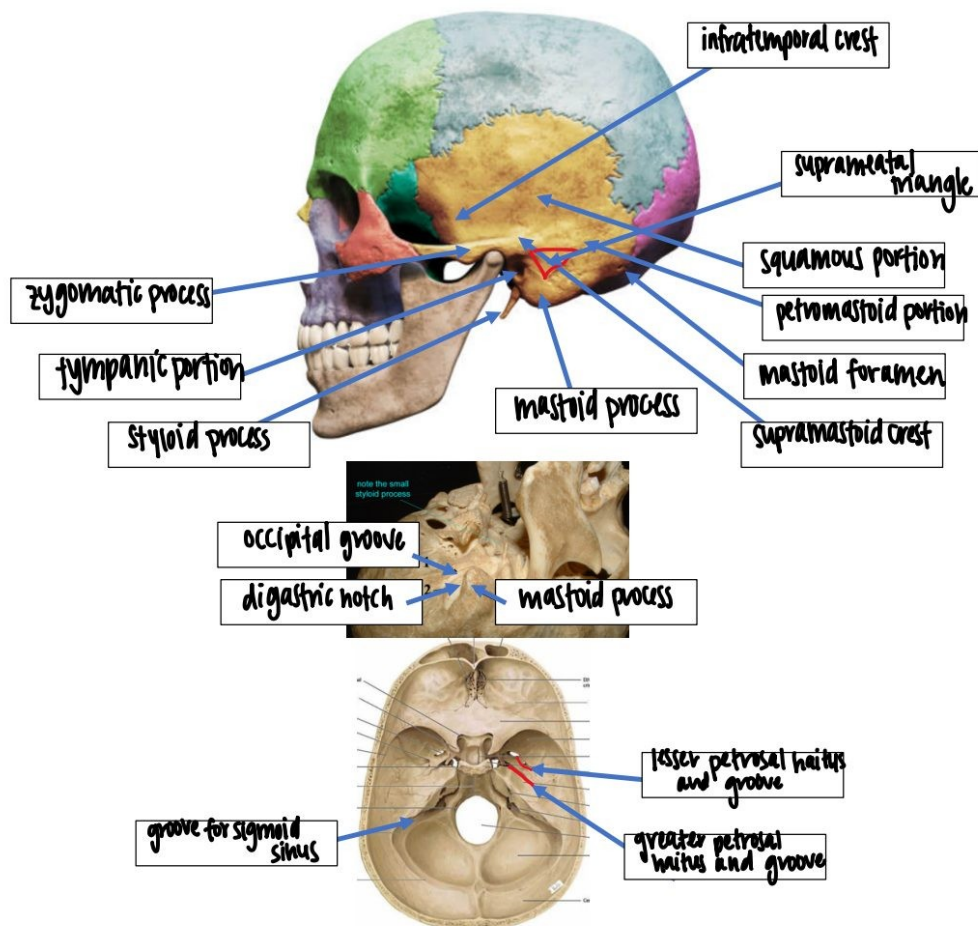
Occipital Bone



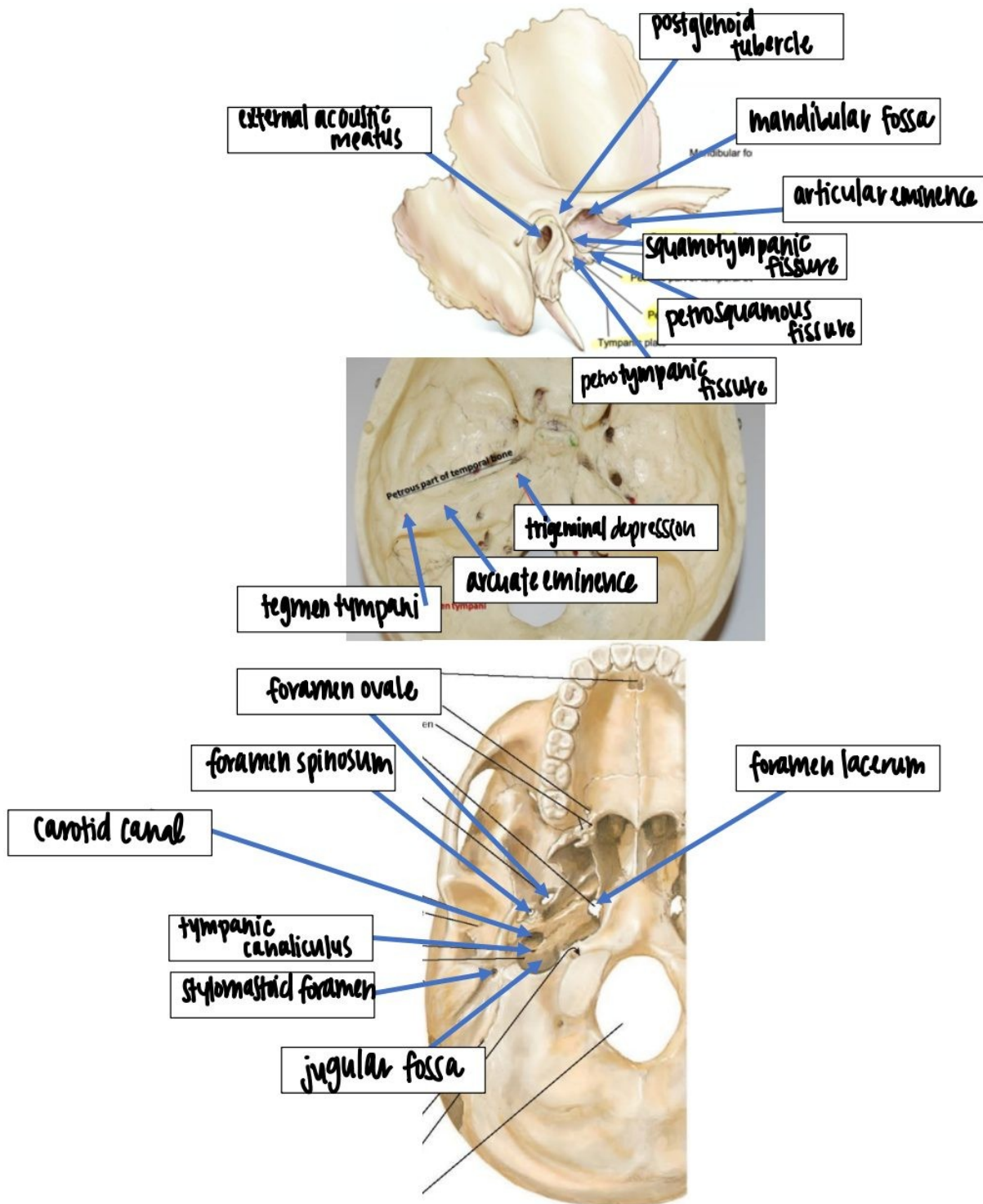
A5. Occipital (internal) and temporal



Temporal Bone

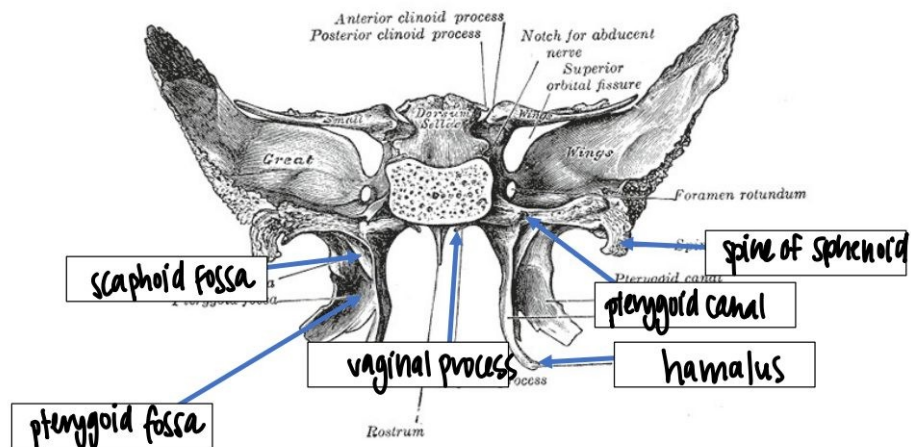
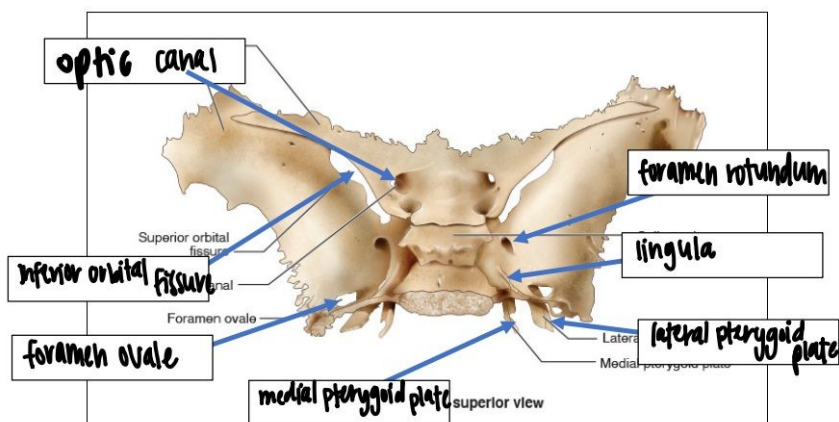
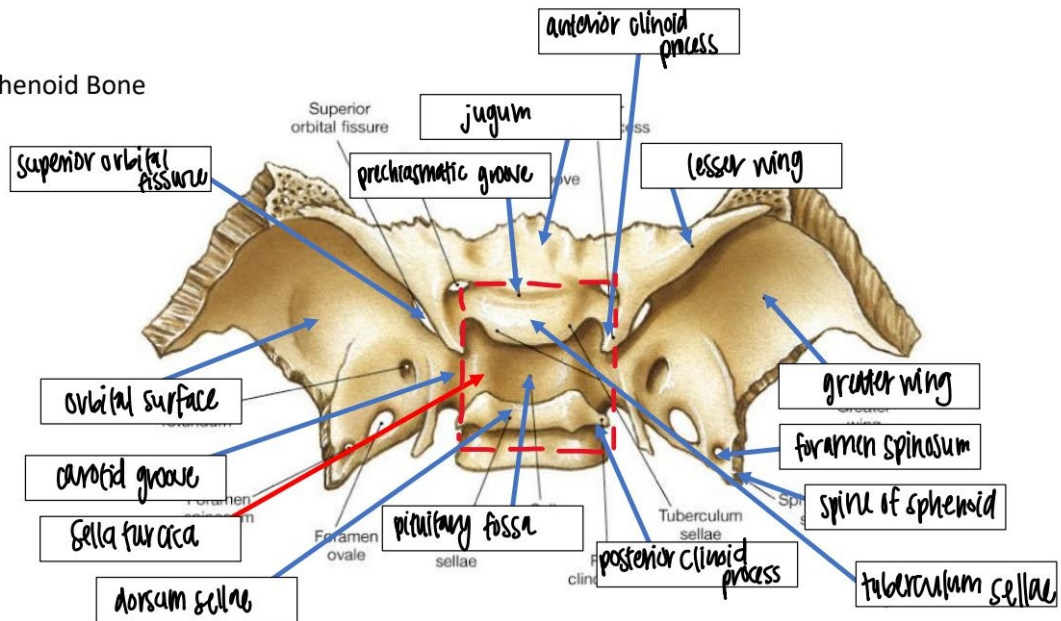


A6. Temporal: TMJ region, petrous part, foramina

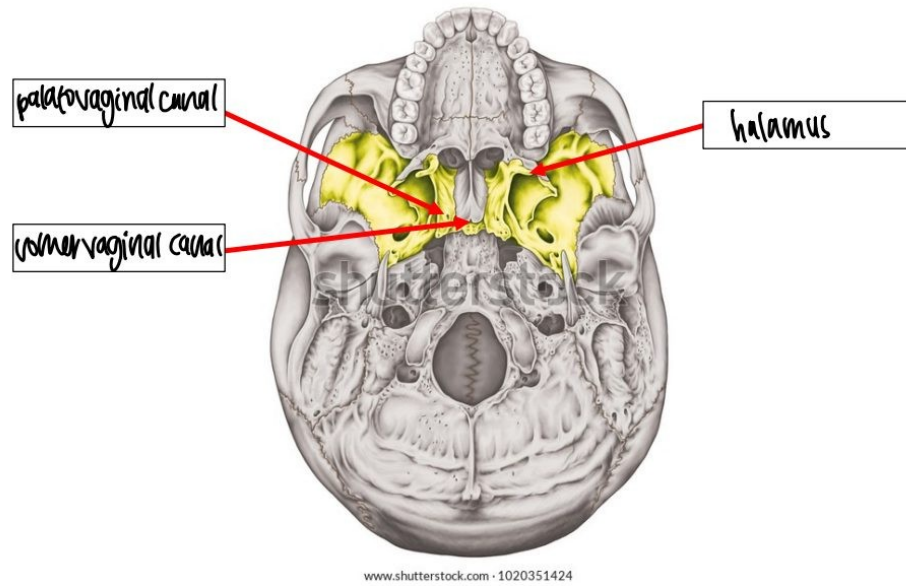


A7. Sphenoid

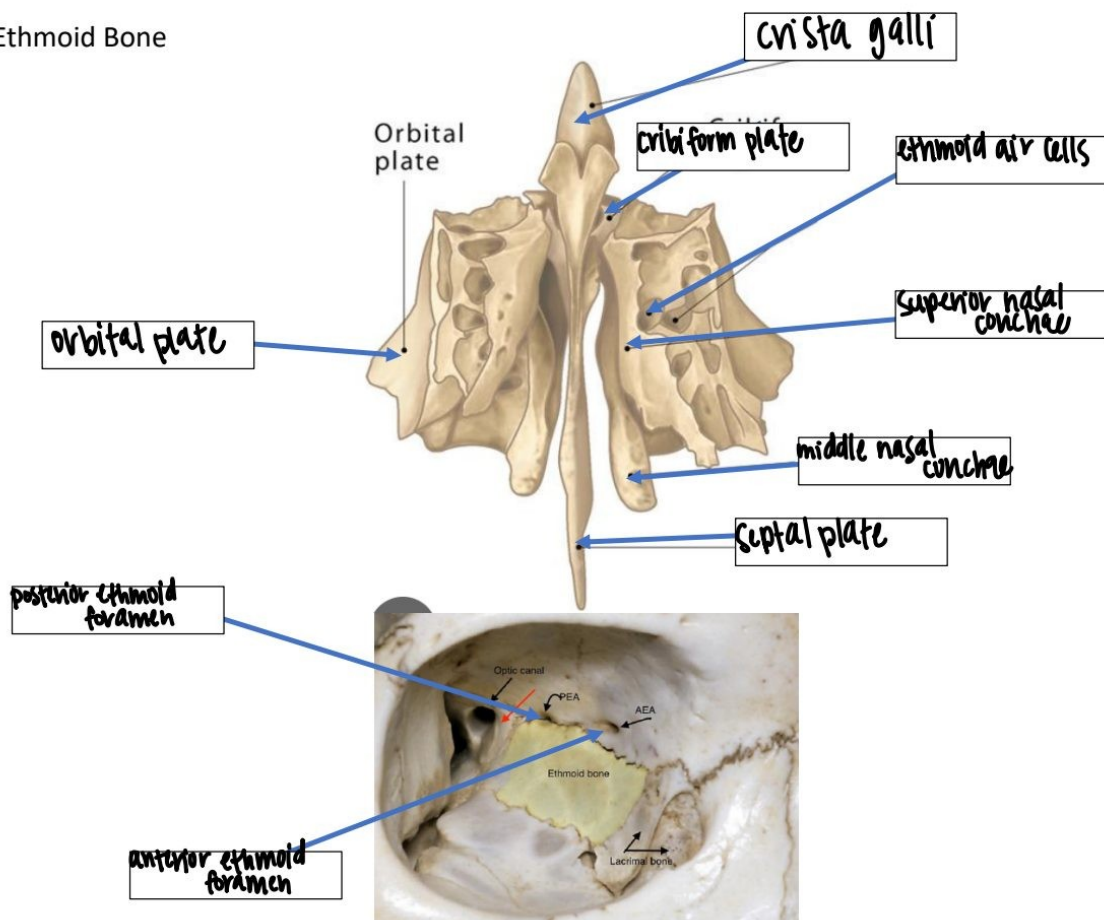
Sphenoid Bone



A8. Pterygoid canals and ethmoid

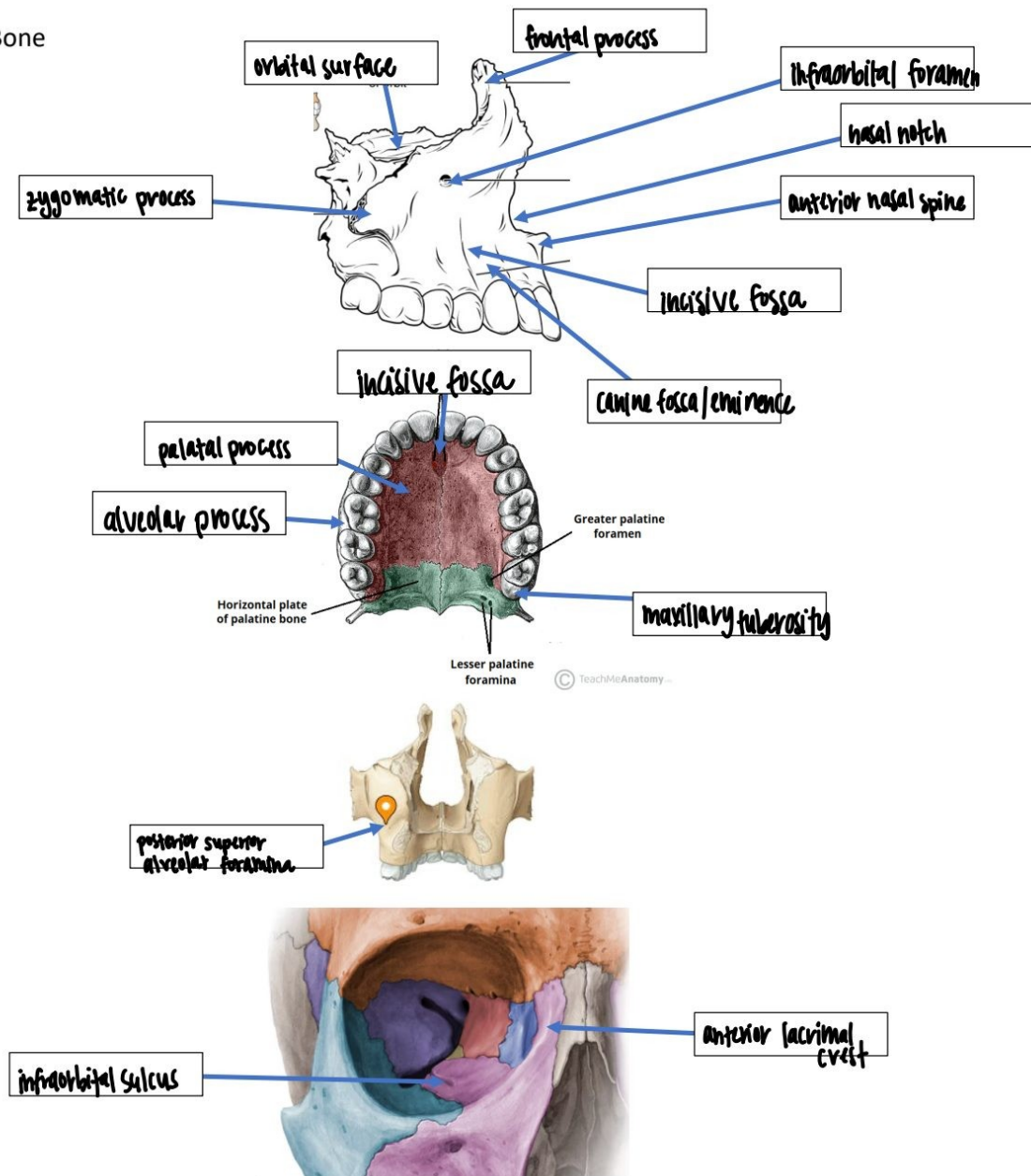


Ethmoid Bone



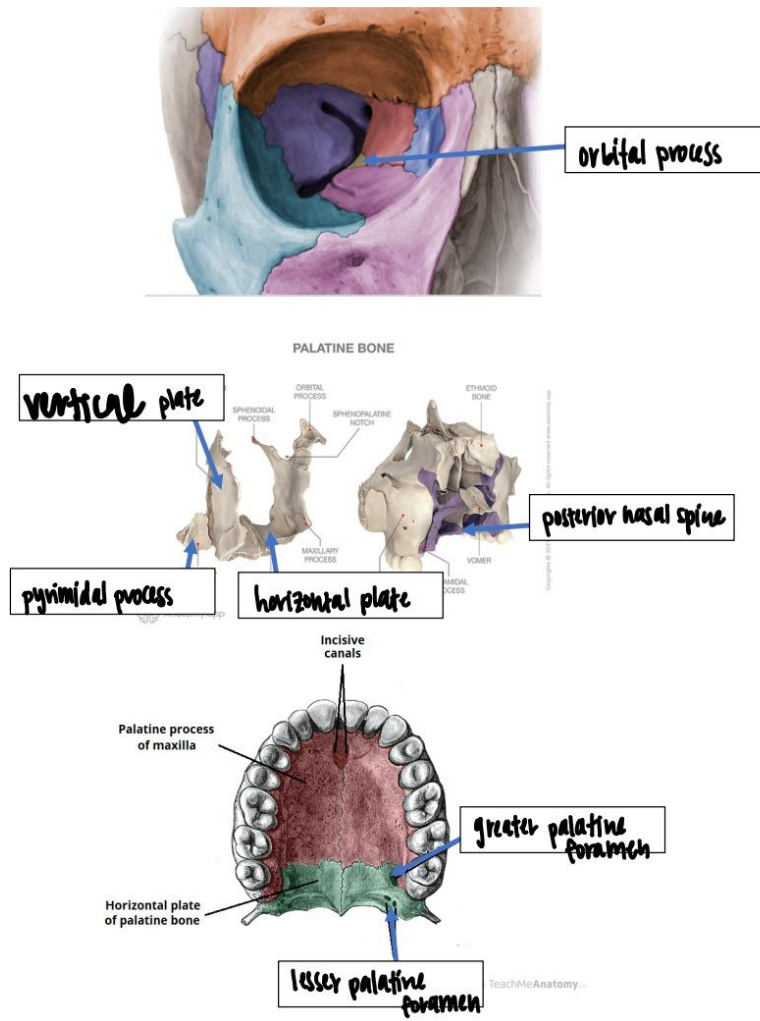
A9. Maxilla

Maxilla Bone

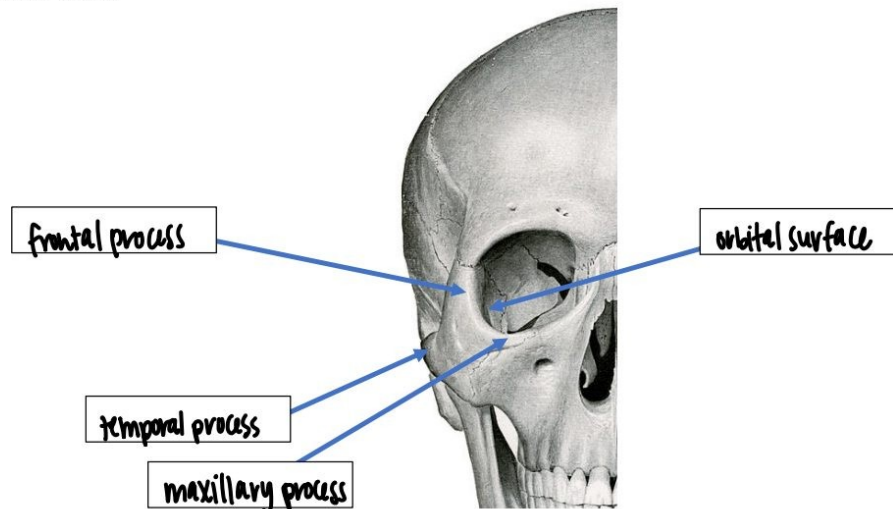


A10. Palatine and zygomatic

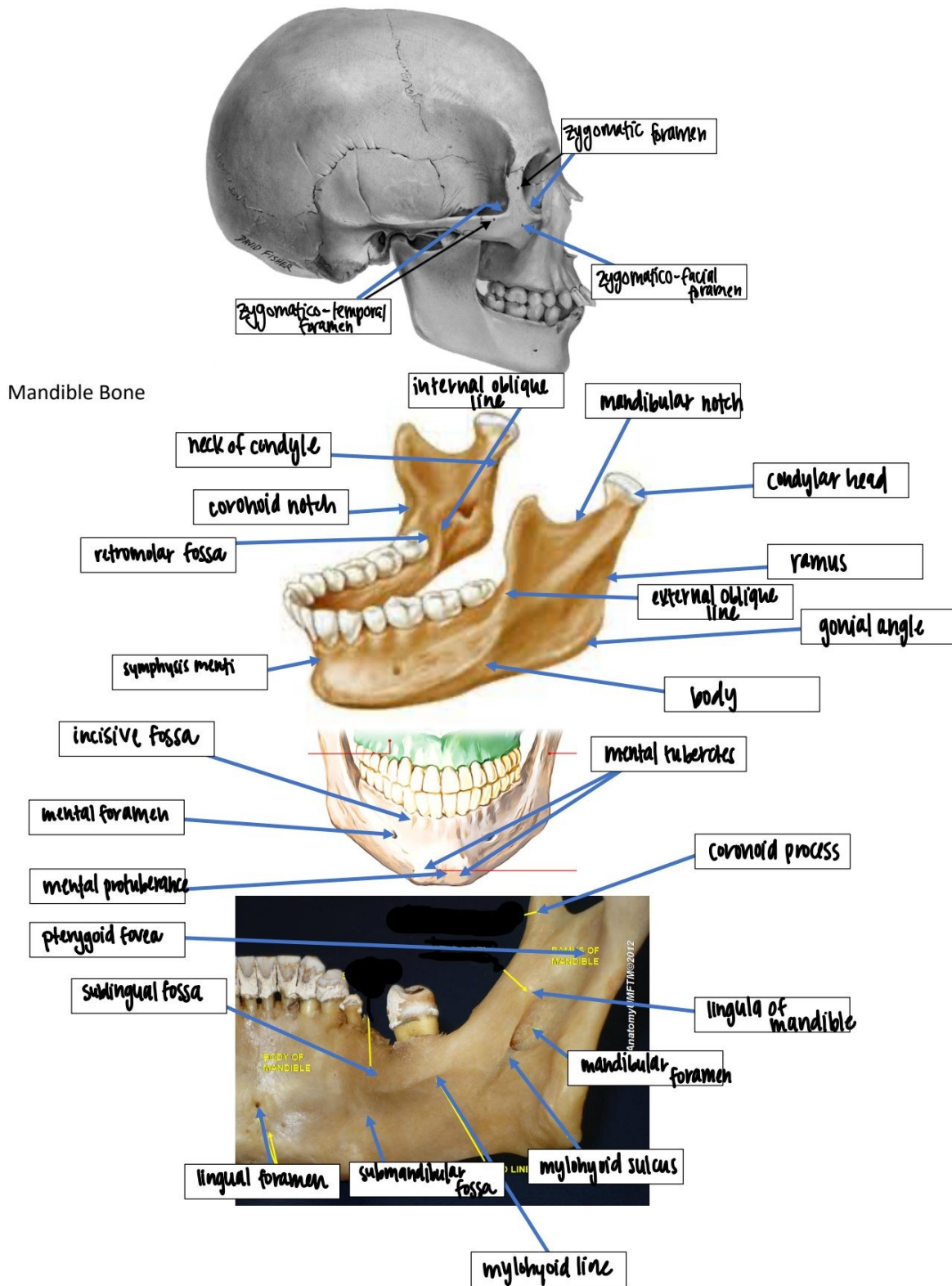
Palatine Bone



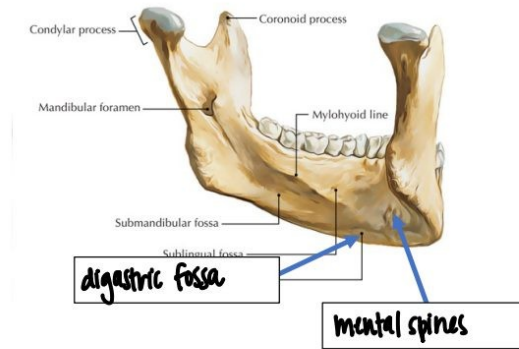
Zygomatic Bone



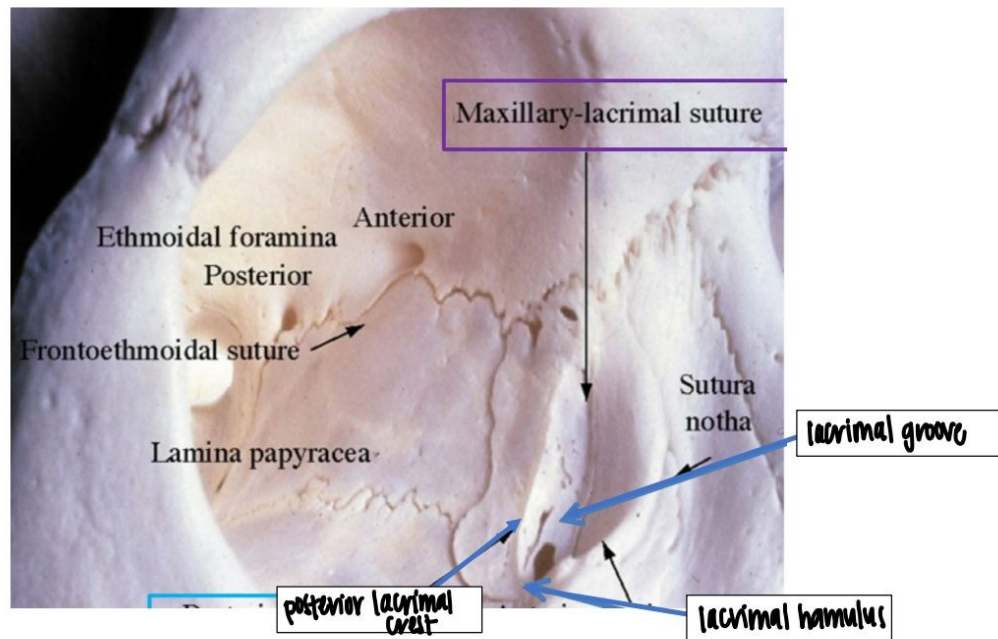
A11. Zygomatic foramina and mandible



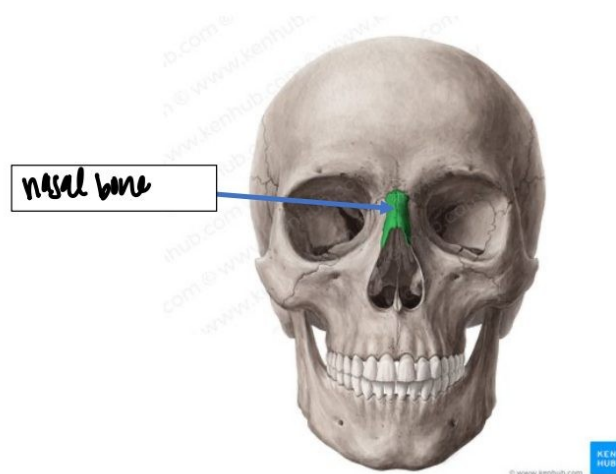
A12. Mandible (lingual), lacrimal, nasal



Lacrimal Bone

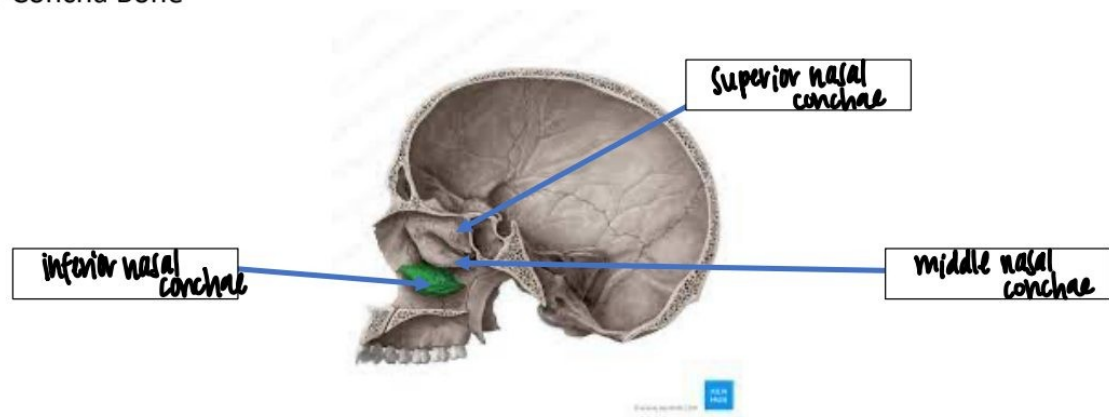


Nasal Bone

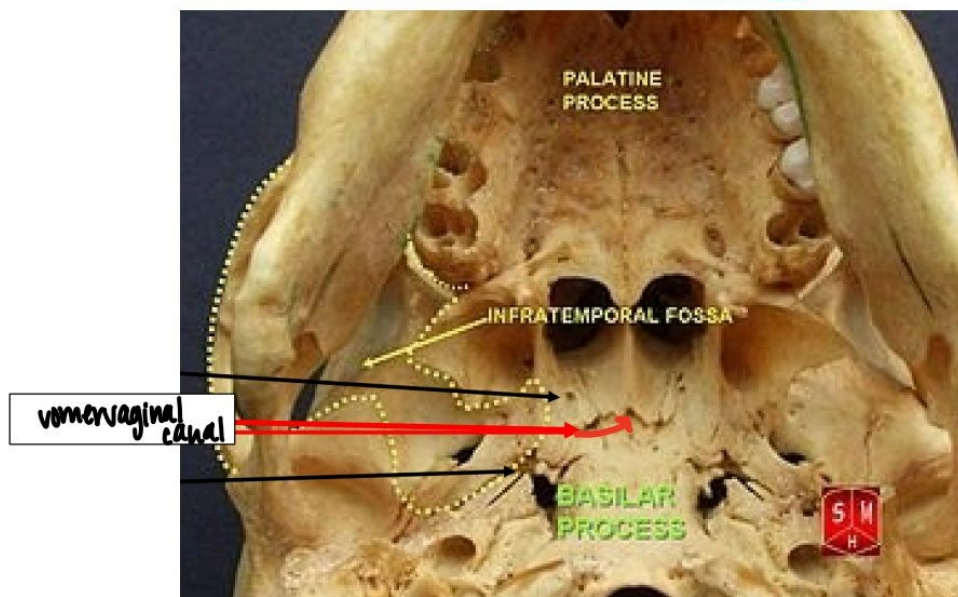
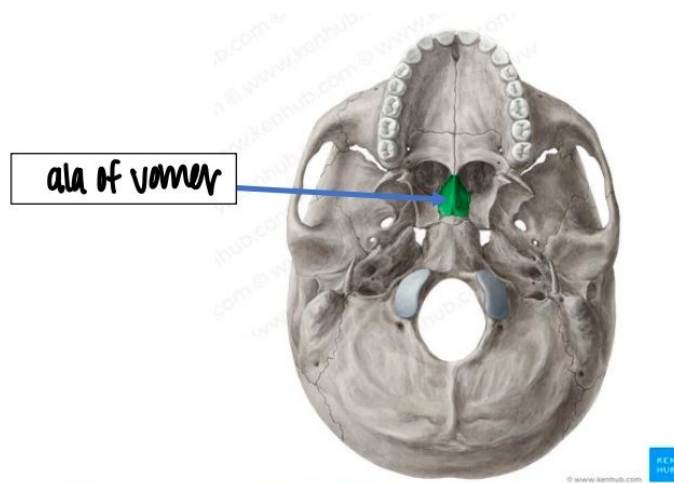


A13. Conchae and vomer

Inferior Concha Bone



Vomer Bone



4. Real-skull photo gallery (P1–P58)

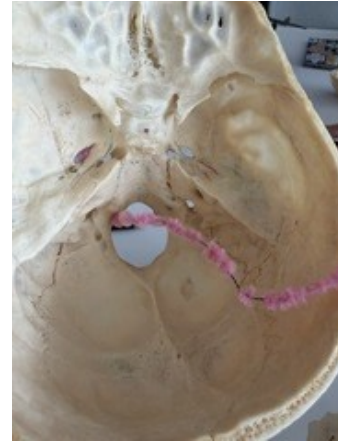
Photos from the High Yield Osteology Quizlet, taken on the lab skulls. The pink pipe cleaner or circle marks the structure.



P1. Scaphoid fossa
Sphenoid



P2. Mylohyoid groove
Mandible



P3. Hypoglossal (anterior condylar) canal
Occipital



P4. Hypoglossal (anterior condylar) canal
Occipital



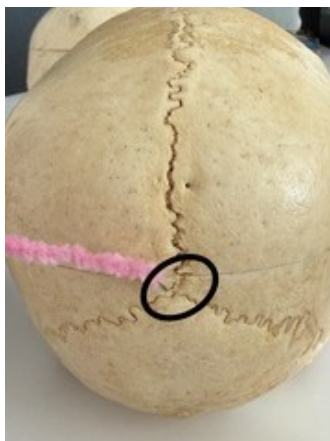
P5. Styloid process
Temporal



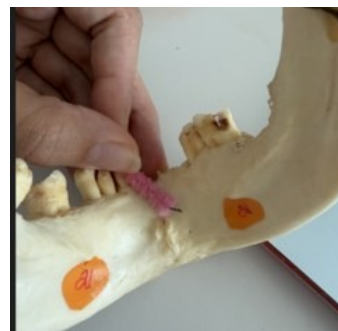
P6. Lingula
Mandible



P7. Carotid canal
Temporal



P8. Lambda
Parietal, occipital



P9. Mental spine
Mandible



P10. Mylohyoid line
Mandible



P11. Petrotympanic fissure
Temporal



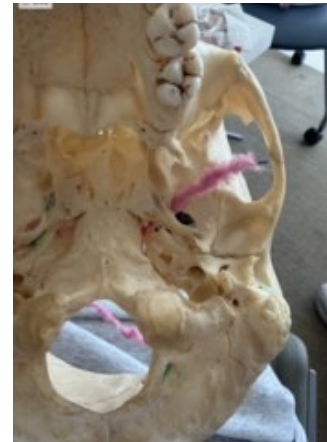
P12. Pterygoid fovea
Mandible



P13. Pterygoid canal
Sphenoid



P14. Foramen ovale
Sphenoid



P15. Foramen ovale
Sphenoid



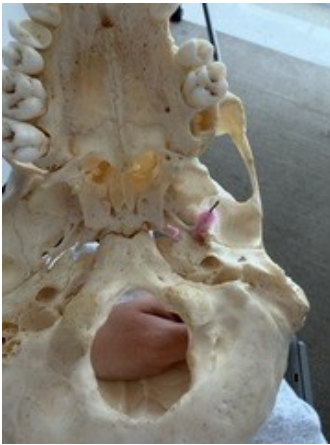
P16. Groove for middle meningeal artery
Parietal or temporal



P17. Digastric notch
Temporal



P18. Foramen spinosum
Sphenoid



P19. Foramen spinosum
Sphenoid



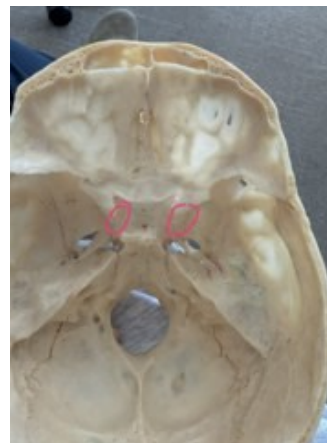
P20. Pterygoid hamulus
Sphenoid



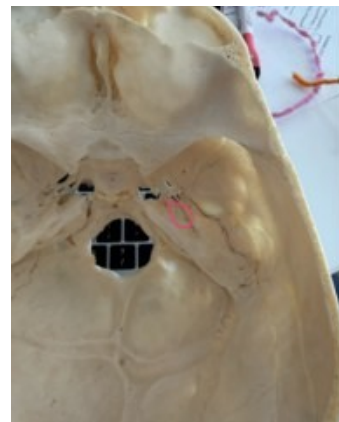
P21. Groove for sigmoid sinus
Temporal



P22. Greater palatine foramen
Palatine



P23. Anterior clinoid process
Sphenoid



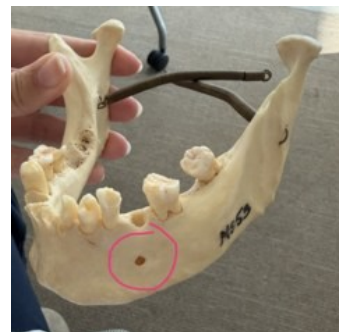
P24. Groove for greater petrosal nerve
Temporal



P25. Groove for lesser petrosal nerve
Temporal



P26. Lateral pterygoid plate
Sphenoid



P27. Mental foramen
Mandible



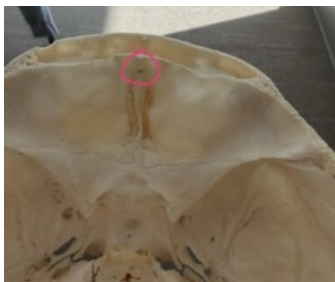
P28. Stylomastoid foramen
Temporal



P29. Internal occipital crest
Occipital



P30. Digastric fossa
Mandible



P31. Foramen cecum
Frontal



P32. Groove for auditory tube
Temporal



P33. Prechiasmatic groove
Sphenoid



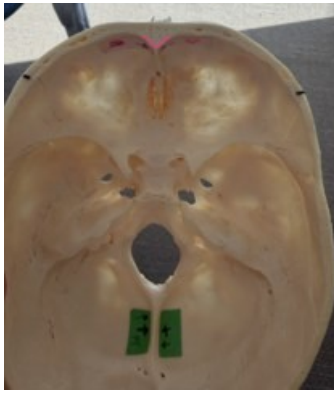
P34. Posterior superior alveolar foramen
Maxilla



P35. Sphenoid spine
Sphenoid



P36. Anterior ethmoidal foramen
Ethmoid, frontal



P37. Frontal crest
Frontal



P38. Inferior temporal line
Parietal



P39. Infraorbital foramen
Maxilla



P40. Trigeminal depression
Temporal



P41. Coronoid process
Mandible



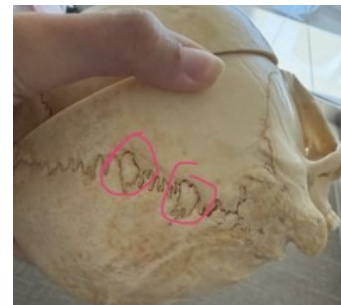
P42. Palatovaginal canal
Palatine, sphenoid



P43. External oblique line
Mandible



P44. Sublingual fossa
Mandible



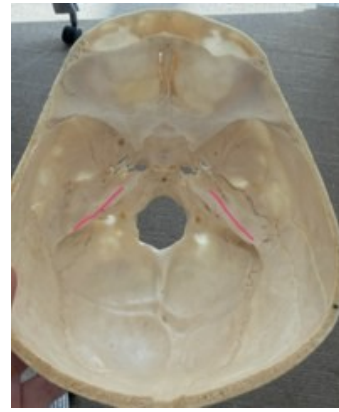
P45. Wormian bones
Varies (parietal/occipital/temporal)



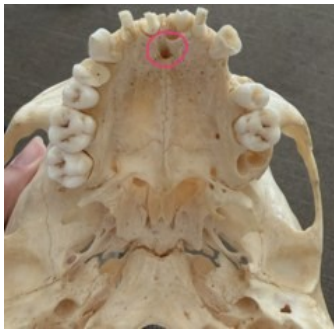
P46. Anterior nasal spine
Maxilla



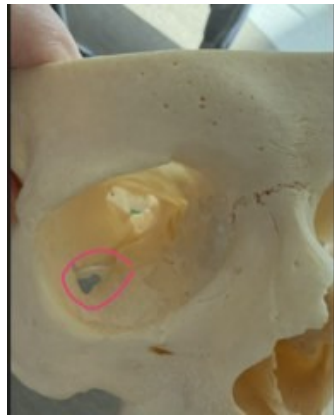
P47. Depressions (foveae) for arachnoid granulations
Parietal/frontal (varies)



P48. Groove for superior petrosal sinus
Temporal



P49. Incisive foramen
Maxilla



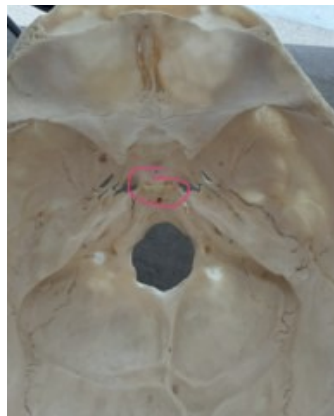
P50. Inferior orbital fissure
Sphenoid, maxilla



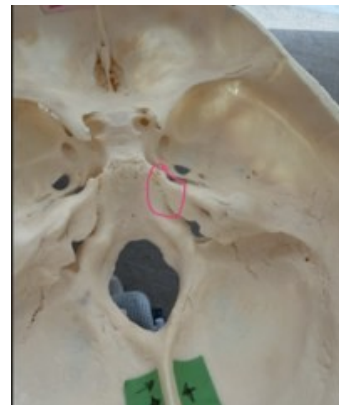
P51. Infraorbital groove
Maxilla



P52. Internal acoustic meatus
Temporal



P53. Dorsum sellae
Sphenoid



P54. Groove for inferior petrosal sinus
Temporal



P55. Zygomaticotemporal foramen
Zygomatic



P56. Zygomaticotemporal foramen
Zygomatic



P57. Nasolacrimal canal
Lacrimal, maxilla



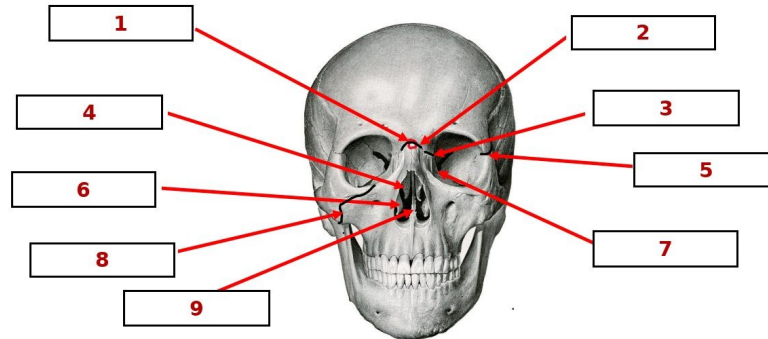
P58. Postglenoid tubercle
Temporal

5. Recall bank

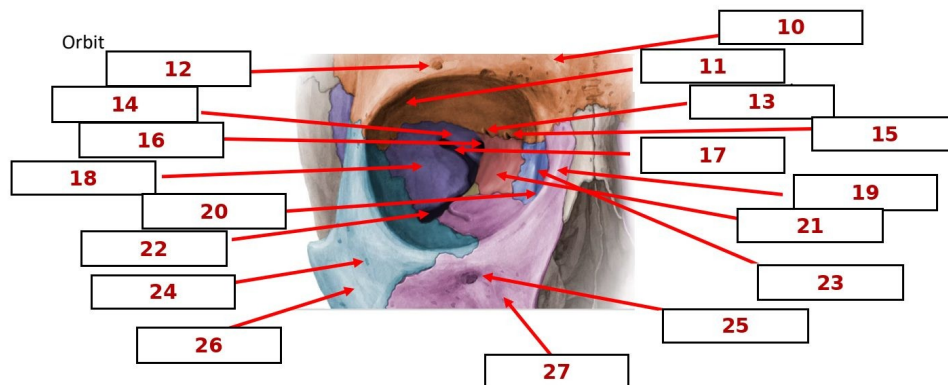
Name each numbered structure, then its bone. Pick your answers from the word bank under each page; some terms are used more than once. Answers are in section 6.

R1. Skull frontal view, orbit, nose (34 labels)

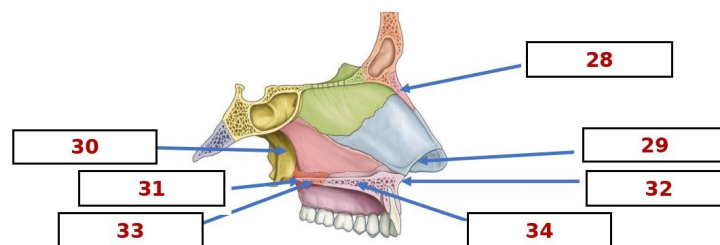
Skull Frontal View



Orbit



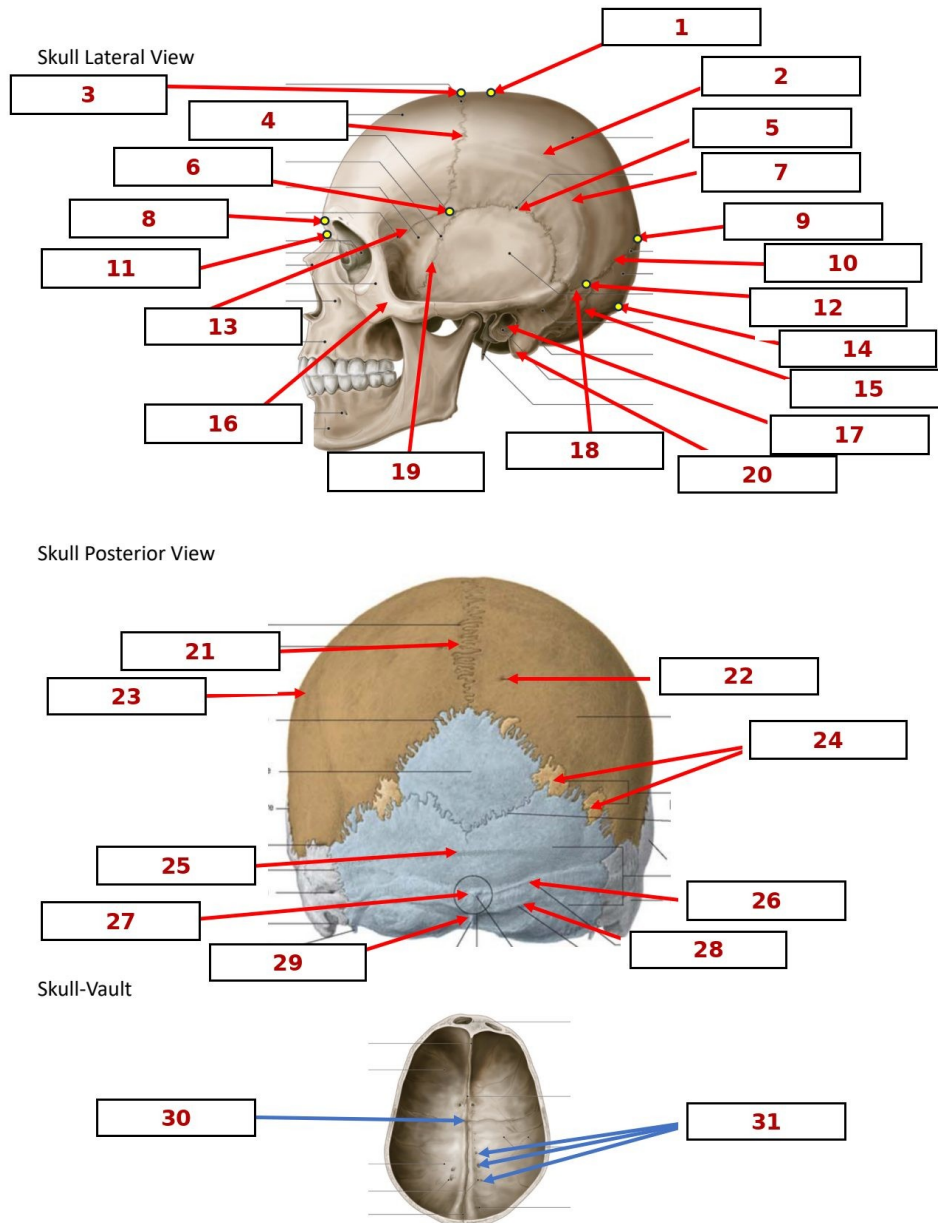
Nose



Word bank

- | | | |
|---|--|--|
| ☐ anterior ethmoidal foramen | ☐ greater wing of sphenoid | ☐ palatal process of maxilla |
| ☐ anterior lacrimal crest | ☐ horizontal plate of palatine bone | ☐ posterior ethmoidal foramen |
| ☐ anterior nasal aperture | ☐ inferior nasal concha | ☐ posterior lacrimal crest |
| ☐ anterior nasal spine | ☐ inferior orbital fissure | ☐ posterior nasal aperture |
| ☐ ethmoid bone | ☐ infraorbital foramen | ☐ posterior nasal spine |
| ☐ fossa for lacrimal gland | ☐ lesser wing of sphenoid | ☐ superior orbital fissure |
| ☐ fossa for lacrimal sac | ☐ maxillary bone | ☐ supraorbital notch |
| ☐ frontal bone | ☐ middle nasal concha | ☐ vomer |
| ☐ frontomaxillary suture | ☐ nasal bones | ☐ zygomatic bone |
| ☐ frontonasal suture | ☐ nasion | ☐ zygomaticofacial foramen |
| ☐ frontozygomatic suture | ☐ optic canal | ☐ zygomaticomaxillary suture |

R2. Lateral view, posterior view, vault (31 labels)

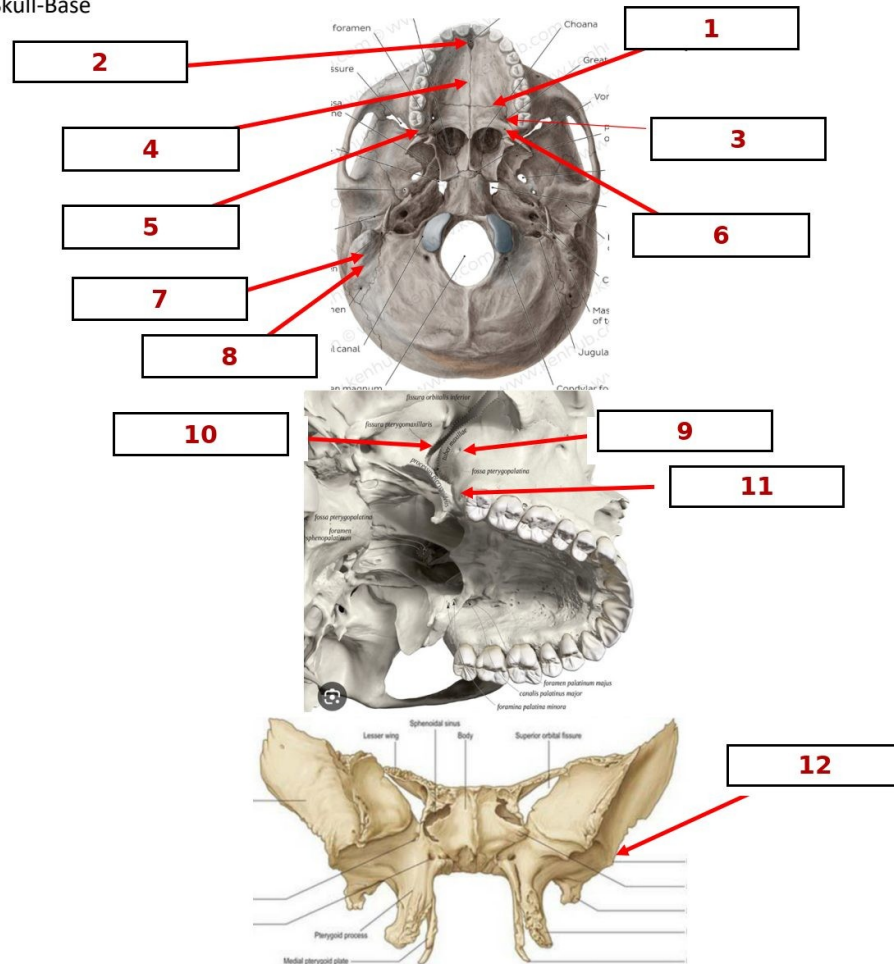


Word bank

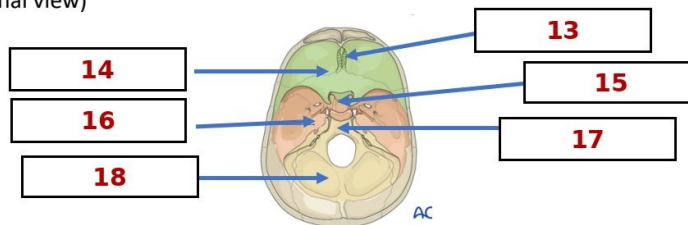
- | | | |
|---|---|--|
| ☐ asterion | ☐ inion | ☐ sagittal suture |
| ☐ bregma | ☐ lambda | ☐ sphenofrontal suture |
| ☐ coronal suture | ☐ lambdoid suture | ☐ sphenosquamosal suture |
| ☐ depressions for arachnoid granulations | ☐ mastoid process | ☐ squamous suture |
| ☐ external occipital crest | ☐ nasion | ☐ superior nuchal line |
| ☐ glabella | ☐ occipitomastoid suture | ☐ superior temporal line |
| ☐ groove for superior sagittal sinus | ☐ parietal eminence | ☐ tympanic portion of temporal bone |
| ☐ highest nuchal line | ☐ parietal foramina | ☐ vertex |
| ☐ inferior nuchal line | ☐ parietomastoid suture | ☐ wormian bones |
| ☐ inferior temporal line | ☐ pterion | ☐ zygomatic arch |

R3. Skull base and cranial fossae (18 labels)

Skull-Base



Regions (internal view)

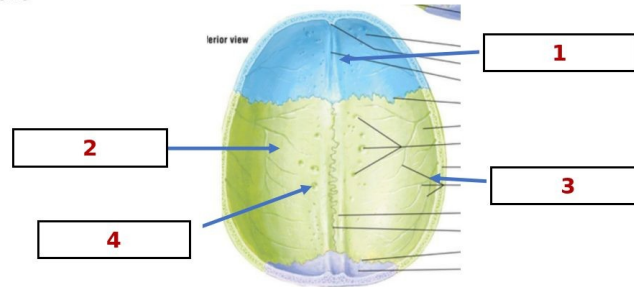


Word bank

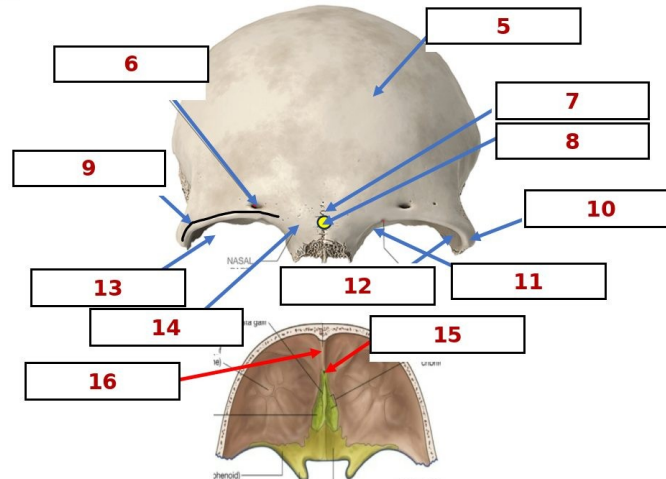
- | | | |
|---|--|--|
| ☐ anterior cranial fossa | ☐ lesser palatine foramen | ☐ posterior cranial fossa |
| ☐ clivus | ☐ mastoid notch | ☐ pterygomaxillary fissure |
| ☐ cribriform plate | ☐ maxillary tuberosity | ☐ sella turcica |
| ☐ greater palatine foramen | ☐ median palatal suture | ☐ transverse palatal suture |
| ☐ incisive foramen | ☐ middle cranial fossa | |
| ☐ infratemporal crest | ☐ occipital groove | |

R4. Parietal, frontal, occipital (external) (24 labels)

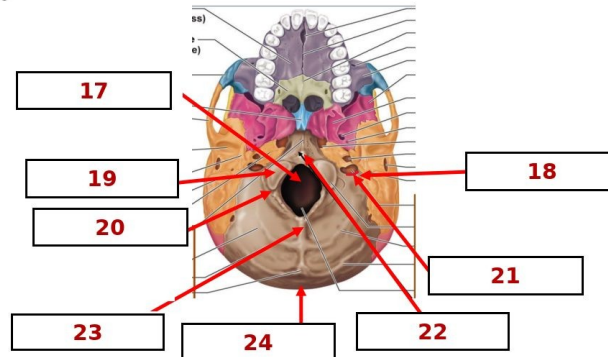
Parietal Bone



Frontal Bone



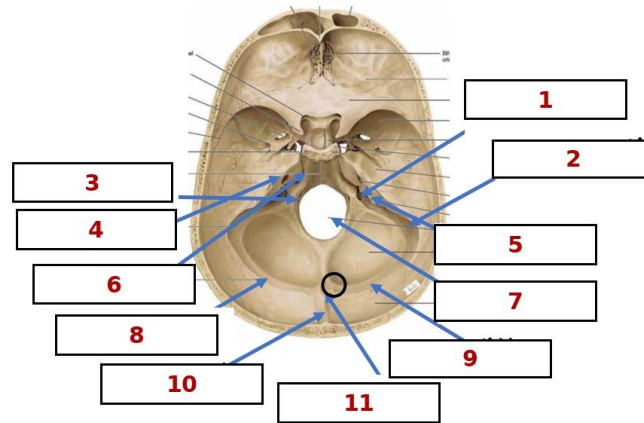
Occipital Bone



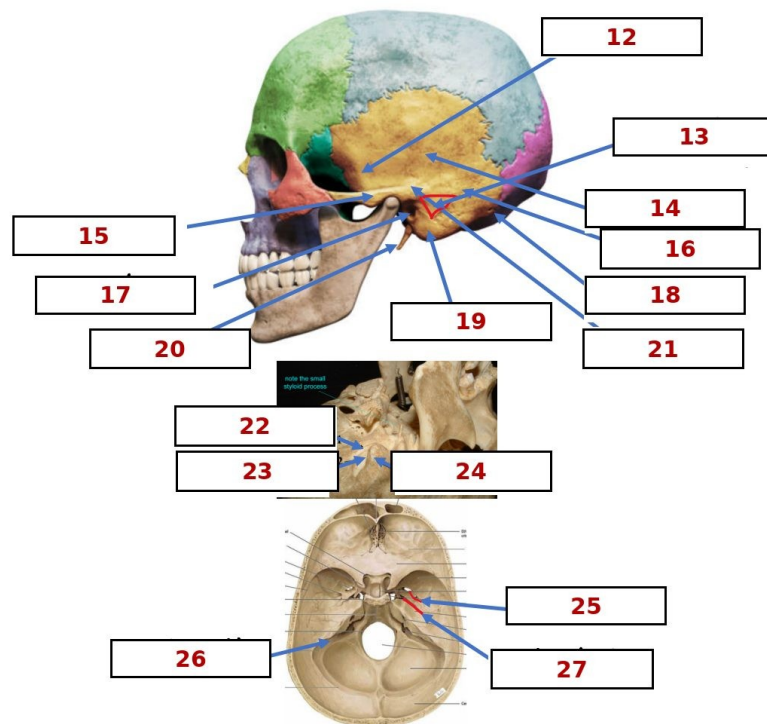
Word bank

- | | | |
|---|--|--|
| ☐ depressions for arachnoid granulations | ☐ grooves for middle meningeal artery | ☐ pharyngeal tubercle |
| ☐ external occipital crest | ☐ jugular notch | ☐ posterior condylar canal |
| ☐ foramen cecum | ☐ jugular process | ☐ squamous portion (frontal) |
| ☐ foramen magnum | ☐ metopic suture | ☐ squamous portion (occipital) |
| ☐ fossa for lacrimal gland | ☐ nasal portion (frontal) | ☐ superciliary arch |
| ☐ frontal crest | ☐ occipital condyle | ☐ supraorbital notch/foramen |
| ☐ glabella | ☐ orbital portion (frontal) | ☐ trochlear fossa |
| ☐ groove for superior sagittal sinus | ☐ parietal bone | ☐ zygomatic process of frontal bone |

R5. Occipital (internal) and temporal (27 labels)



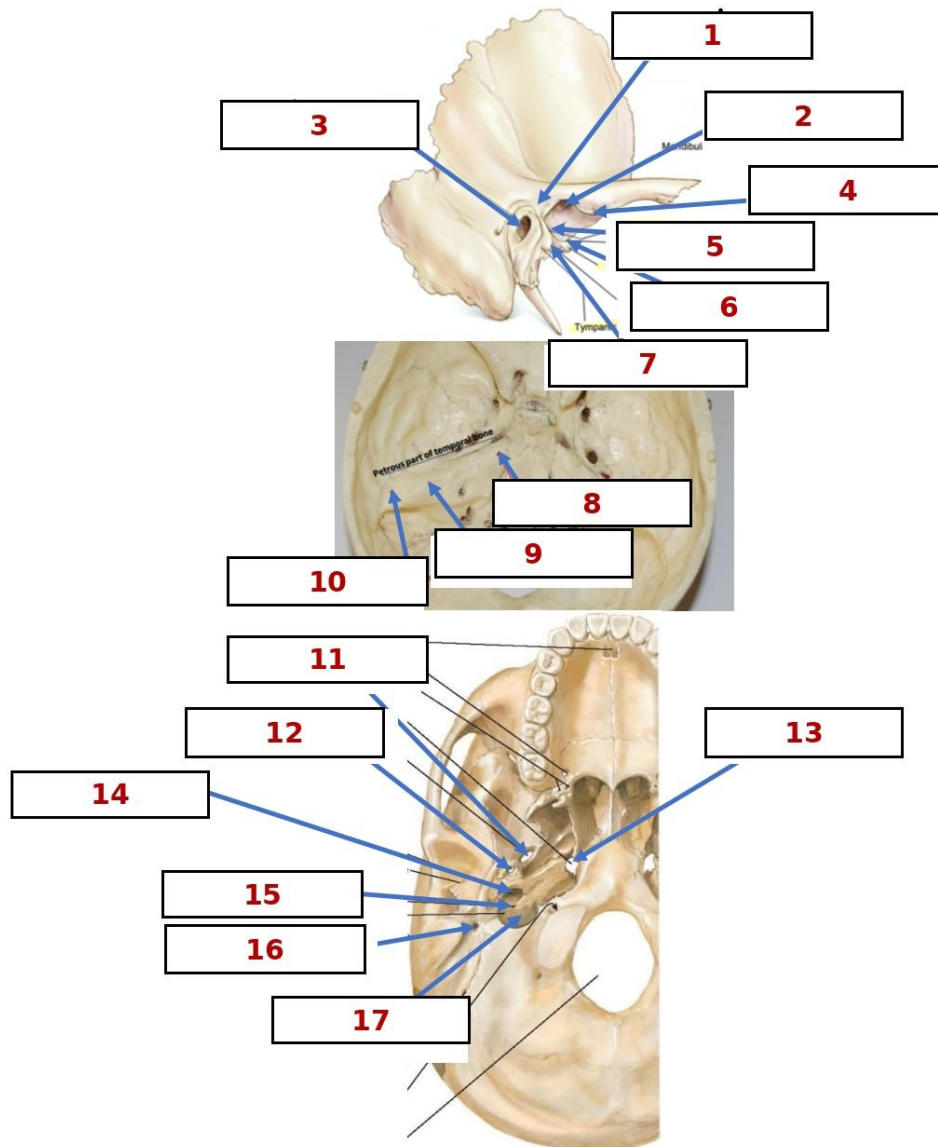
Temporal Bone



Word bank

- | | | |
|---|--|---|
| ☐ digastric notch | ☐ infratemporal crest | ☐ petromastoid portion |
| ☐ foramen magnum | ☐ internal occipital crest | ☐ squamous portion (temporal) |
| ☐ greater petrosal hiatus and groove | ☐ internal occipital protuberance | ☐ styloid process |
| ☐ groove for inferior petrosal sinus | ☐ jugular foramen | ☐ supramastoid crest |
| ☐ groove for left transverse sinus | ☐ jugular tubercle | ☐ suprameatal triangle |
| ☐ groove for right transverse sinus | ☐ lesser petrosal hiatus and groove | ☐ tympanic portion |
| ☐ groove for sigmoid sinus | ☐ mastoid foramen | ☐ zygomatic process (temporal) |
| ☐ groove for superior petrosal sinus | ☐ mastoid process | |
| ☐ hypoglossal canal | ☐ occipital groove | |

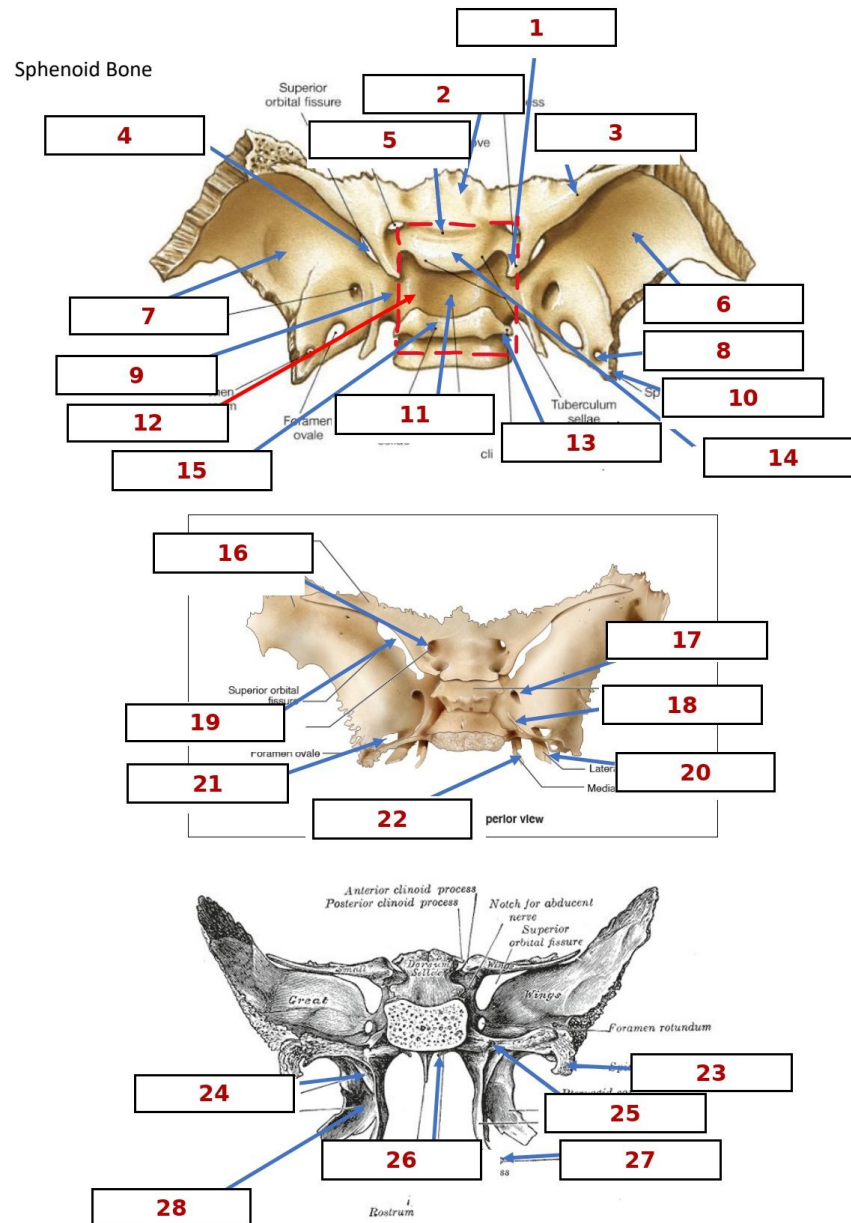
R6. Temporal: TMJ region, petrous part, foramina (17 labels)



Word bank

- | | | |
|---|--|---|
| ☐ arcuate eminence | ☐ foramen spinosum | ☐ squamotympanic fissure |
| ☐ articular eminence | ☐ jugular fossa | ☐ stylomastoid foramen |
| ☐ carotid canal | ☐ mandibular fossa | ☐ tegmen tympani |
| ☐ external acoustic meatus | ☐ petrosquamous fissure | ☐ trigeminal depression |
| ☐ foramen lacerum | ☐ petrotympanic fissure | ☐ tympanic canaliculus |
| ☐ foramen ovale | ☐ postglenoid tubercle | |

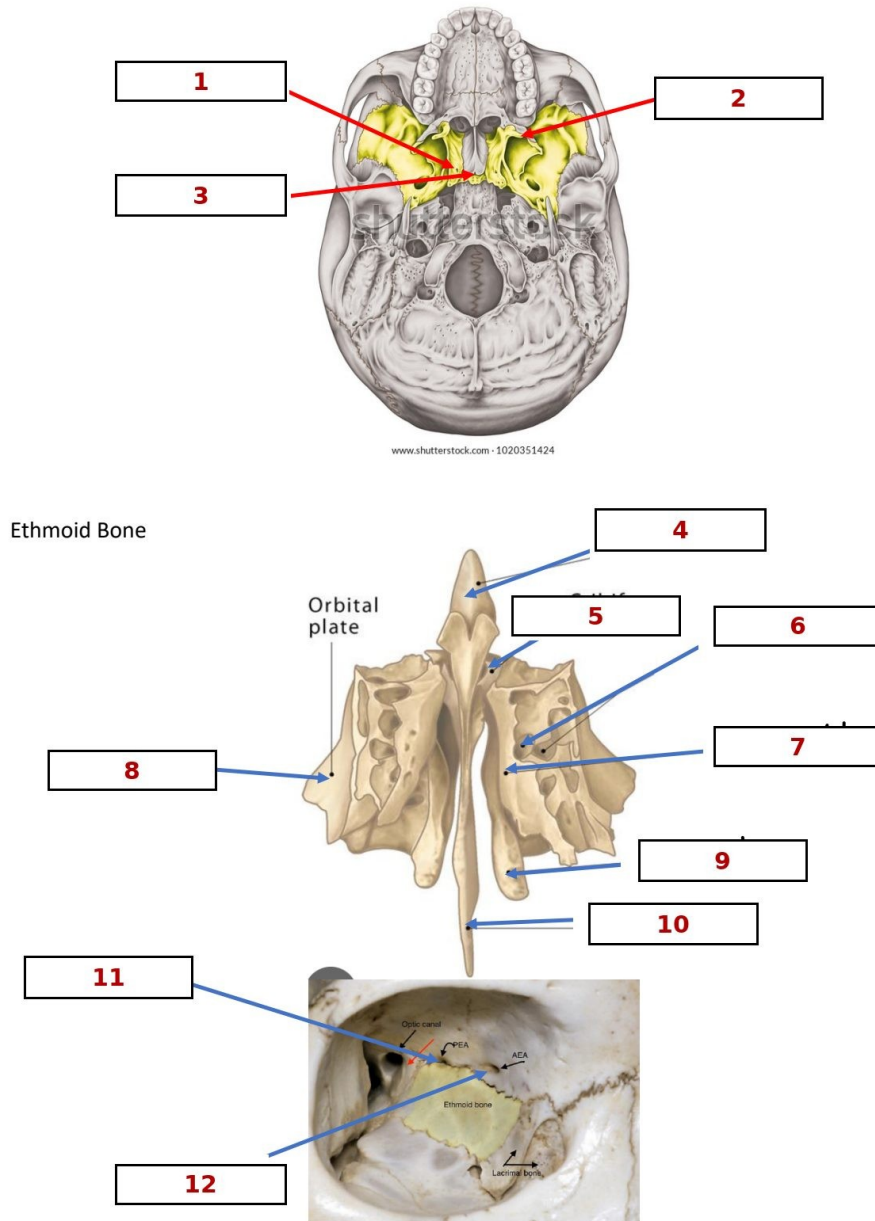
R7. Sphenoid (28 labels)



Word bank

- | | | |
|---|--|---|
| ☐ anterior clinoid process | ☐ lateral pterygoid plate | ☐ pterygoid canal |
| ☐ carotid groove | ☐ lesser wing | ☐ pterygoid fossa |
| ☐ dorsum sellae | ☐ lingula (sphenoid) | ☐ pterygoid hamulus |
| ☐ foramen ovale | ☐ medial pterygoid plate | ☐ scaphoid fossa |
| ☐ foramen rotundum | ☐ optic canal | ☐ sella turcica |
| ☐ foramen spinosum | ☐ orbital surface (sphenoid) | ☐ spine of sphenoid |
| ☐ greater wing | ☐ pituitary (hypophyseal) fossa | ☐ superior orbital fissure |
| ☐ inferior orbital fissure | ☐ posterior clinoid process | ☐ tuberculum sellae |
| ☐ jugum | ☐ prechiasmatic groove | ☐ vaginal process |

R8. Pterygoid canals and ethmoid (12 labels)

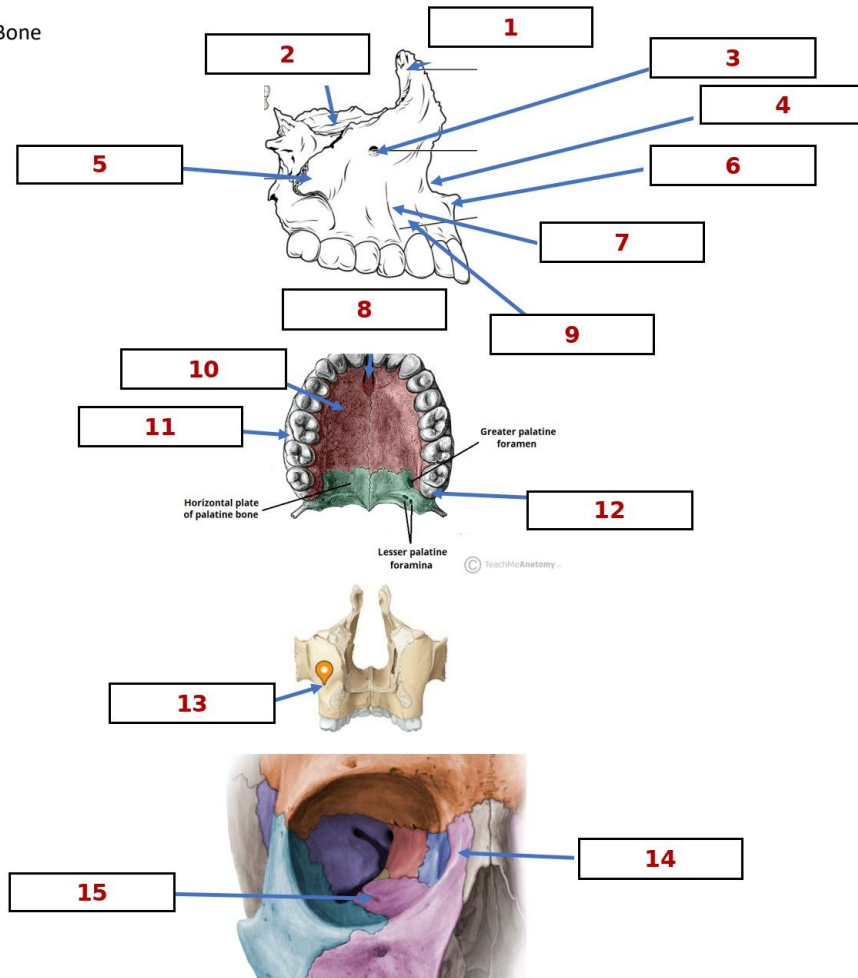


Word bank

- | | | |
|---|---|--|
| ☐ anterior ethmoidal foramen | ☐ middle nasal concha | ☐ posterior ethmoidal foramen |
| ☐ cribriform plate | ☐ orbital plate (ethmoid) | ☐ pterygoid hamulus |
| ☐ crista galli | ☐ palatovaginal canal | ☐ superior nasal concha |
| ☐ ethmoid air cells | ☐ perpendicular (septal) plate | ☐ vomerovaginal canal |

R9. Maxilla (15 labels)

Maxilla Bone

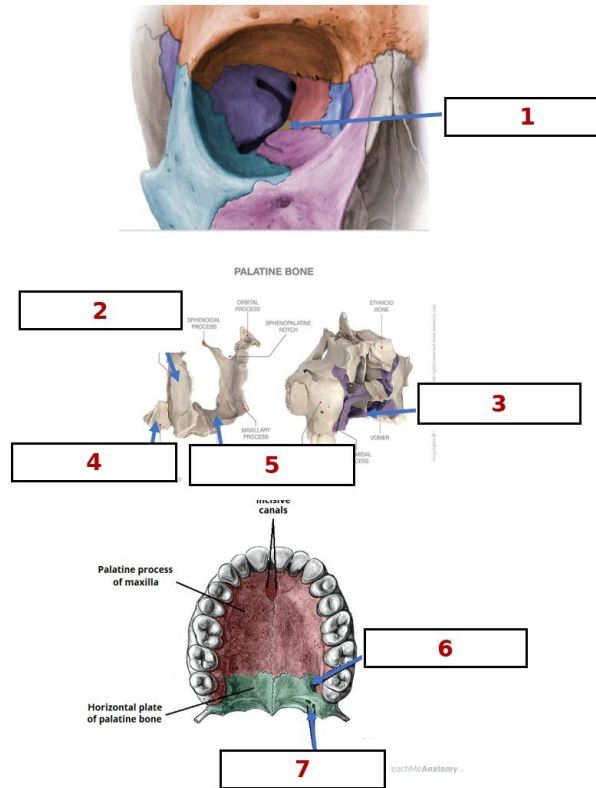


Word bank

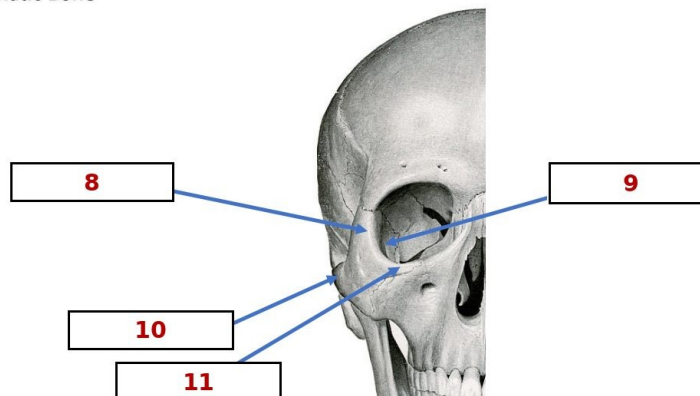
- | | | |
|--|---|---|
| ☐ alveolar process | ☐ incisive fossa | ☐ nasal notch |
| ☐ anterior lacrimal crest | ☐ incisive fossa (palatal) | ☐ orbital surface (maxilla) |
| ☐ anterior nasal spine | ☐ infraorbital foramen | ☐ palatal process |
| ☐ canine fossa/eminence | ☐ infraorbital sulcus (groove) | ☐ posterior superior alveolar foramina |
| ☐ frontal process (maxilla) | ☐ maxillary tuberosity | ☐ zygomatic process (maxilla) |

R10. Palatine and zygomatic (11 labels)

Palatine Bone



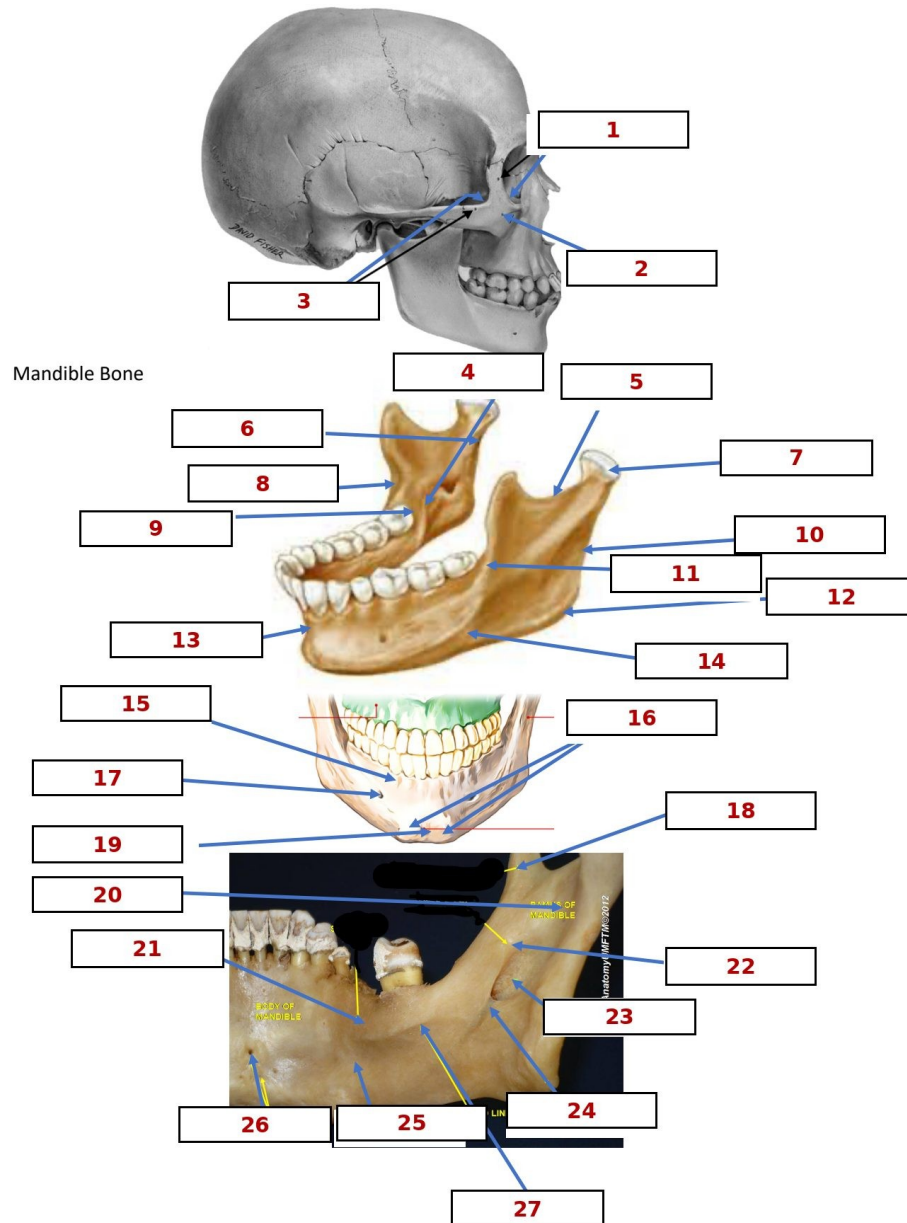
Zygomatic Bone



Word bank

- | | | |
|--|--|--|
| ☐ frontal process (zygomatic) | ☐ maxillary process | ☐ pyramidal process |
| ☐ greater palatine foramen | ☐ orbital process (palatine) | ☐ temporal process |
| ☐ horizontal plate | ☐ orbital surface (zygomatic) | ☐ vertical plate |
| ☐ lesser palatine foramen | ☐ posterior nasal spine | |

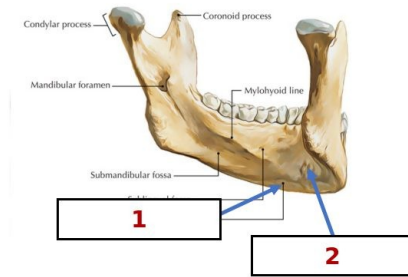
R11. Zygomatic foramina and mandible (27 labels)



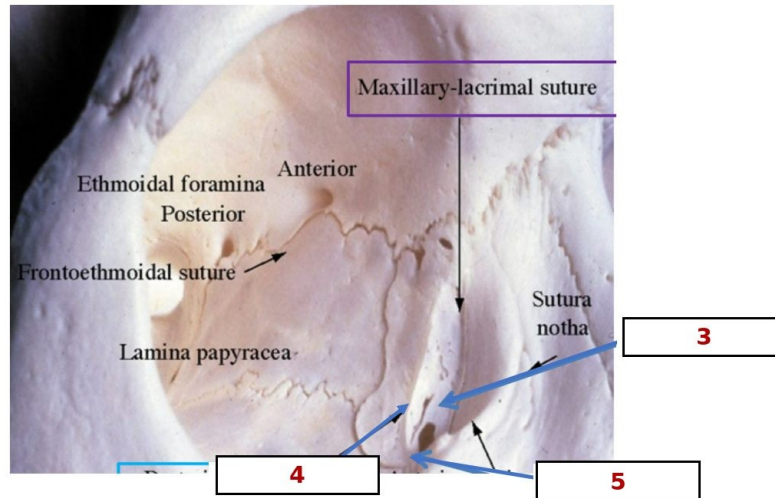
Word bank

- | | | |
|--|--|---|
| ☐ body | ☐ lingula of mandible | ☐ pterygoid fovea |
| ☐ condylar head | ☐ mandibular foramen | ☐ ramus |
| ☐ coronoid notch | ☐ mandibular notch | ☐ retromolar fossa |
| ☐ coronoid process | ☐ mental foramen | ☐ sublingual fossa |
| ☐ external oblique line | ☐ mental protuberance | ☐ submandibular fossa |
| ☐ gonial angle | ☐ mental tubercles | ☐ symphysis menti |
| ☐ incisive fossa (mandible) | ☐ mylohyoid line | ☐ zygomatic foramen |
| ☐ internal oblique line | ☐ mylohyoid sulcus (groove) | ☐ zygomaticofacial foramen |
| ☐ lingual foramen | ☐ neck of condyle | ☐ zygomaticotemporal foramen |

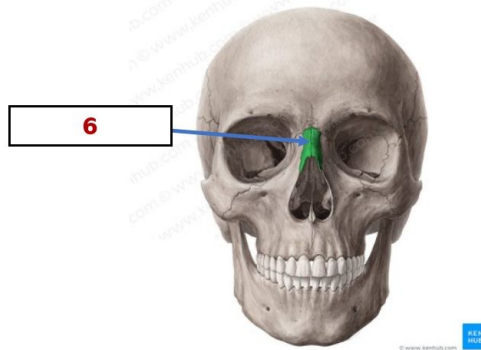
R12. Mandible (lingual), lacrimal, nasal (6 labels)



Lacrimal Bone



Nasal Bone



Word bank

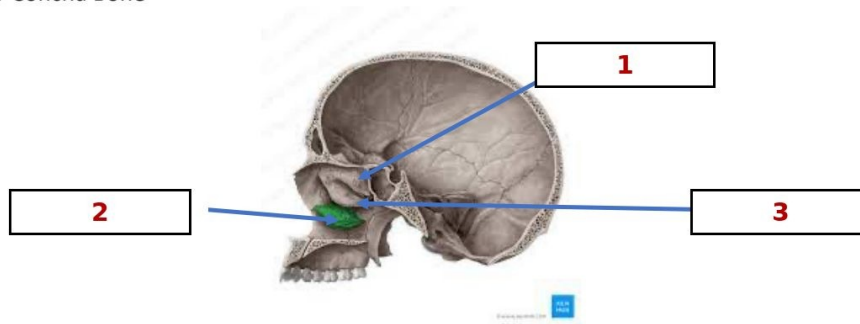
- ☐ digastric fossa
- ☐ lacrimal groove

- ☐ lacrimal hamulus
- ☐ mental spines

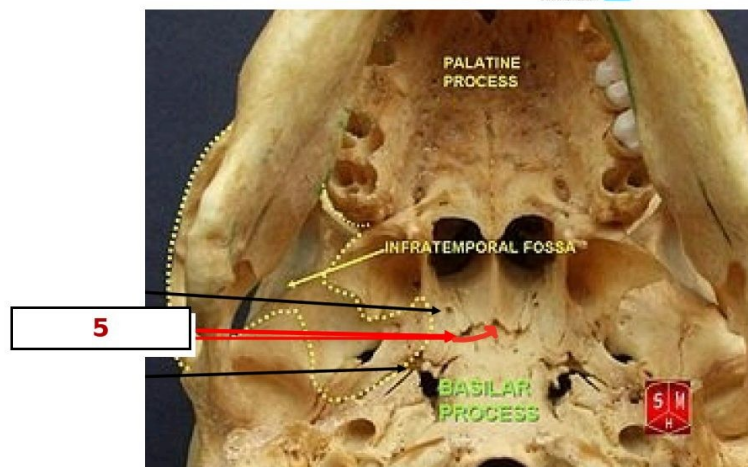
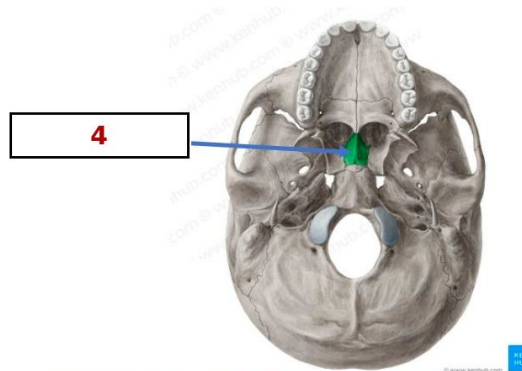
- ☐ nasal bone
- ☐ posterior lacrimal crest

R13. Conchae and vomer (5 labels)

Inferior Concha Bone



Vomer Bone



Word bank

- ☐ ala of vomer
- ☐ inferior nasal concha

- ☐ middle nasal concha
- ☐ superior nasal concha

- ☐ vomerovaginal canal

Real-skull photo recall (Q1–Q58)

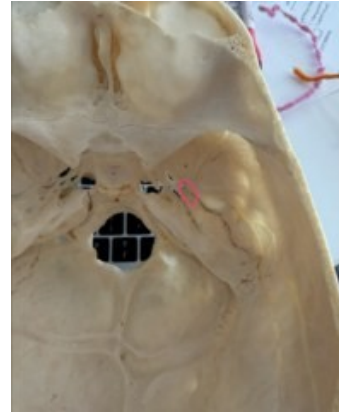
Same photos as the gallery, shuffled. Name the marked structure and its bone.



Q1



Q2



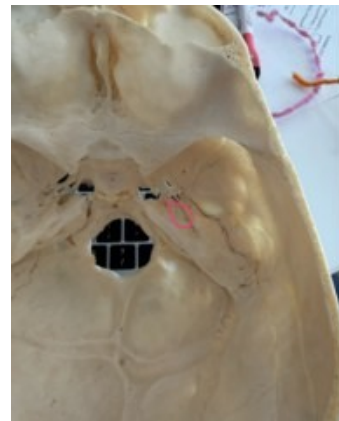
Q3



Q4



Q5



Q6



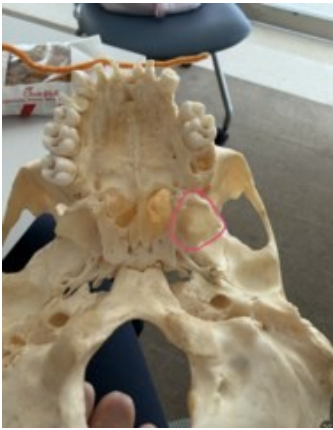
Q7



Q8



Q9



Q10



Q11



Q12



Q13



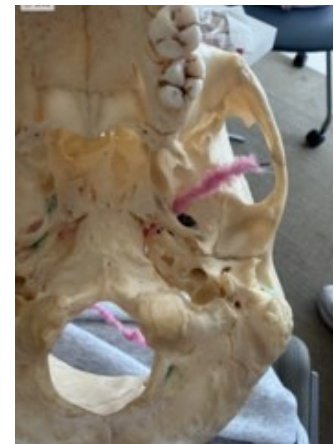
Q14



Q15



Q16



Q17



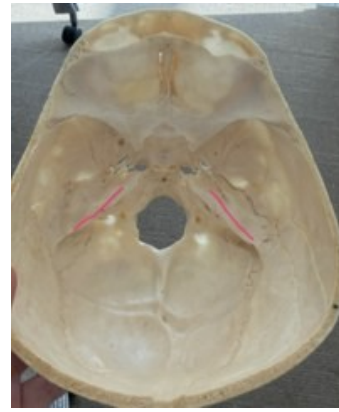
Q18



Q19



Q20



Q21



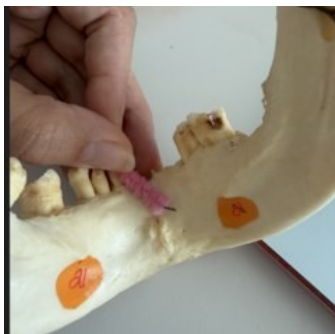
Q22



Q23



Q24



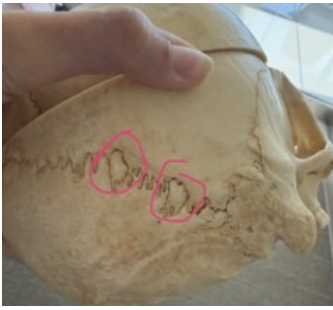
Q25



Q26



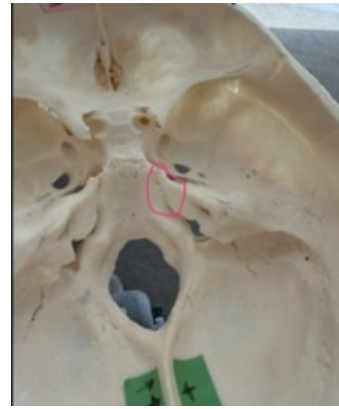
Q27



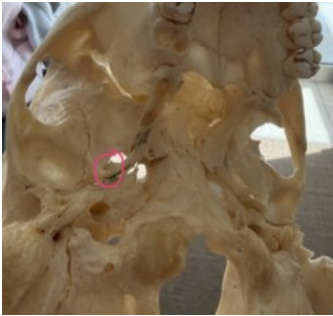
Q28



Q29



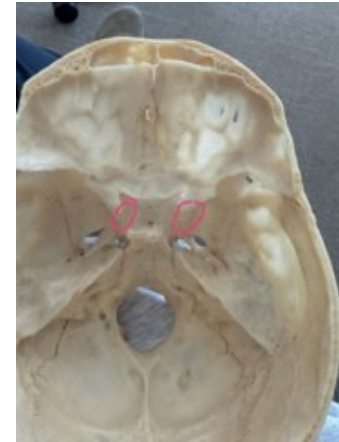
Q30



Q31



Q32



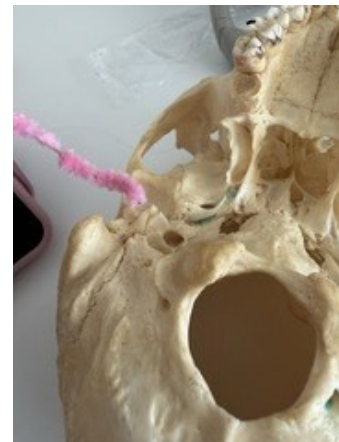
Q33



Q34



Q35



Q36



Q37



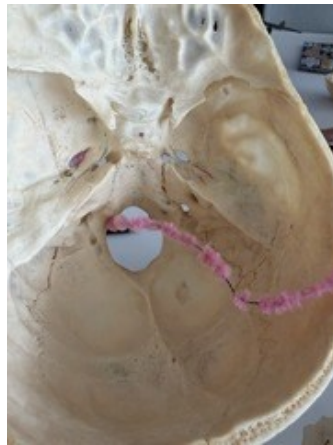
Q38



Q39



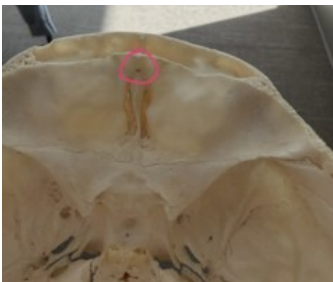
Q40



Q41



Q42



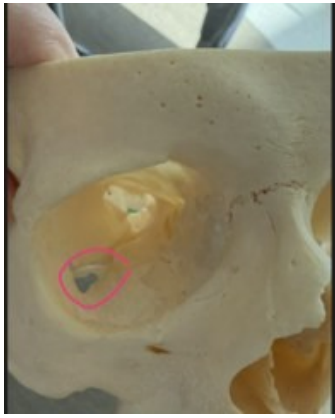
Q43



Q44



Q45



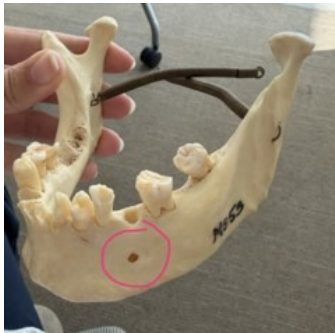
Q46



Q47



Q48



Q49



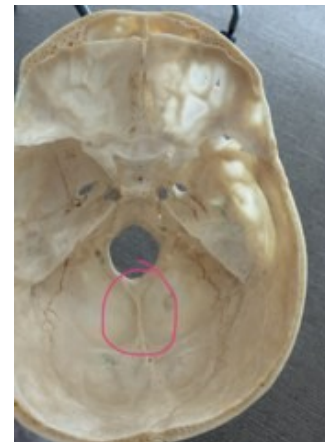
Q50



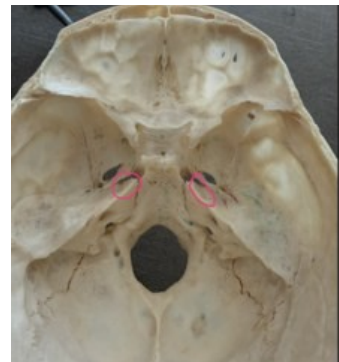
Q51



Q52



Q53



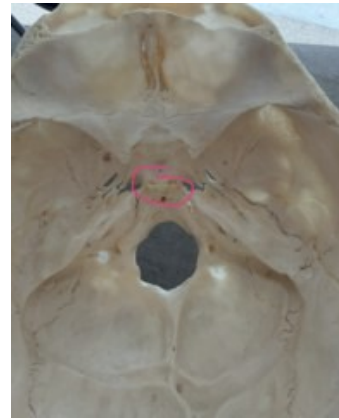
Q54



Q55



Q56



Q57



Q58

Word bank

- | | | |
|--|--|--|
| ☐ Anterior clinoid process | ☐ Groove for lesser petrosal nerve | ☐ Mylohyoid line |
| ☐ Anterior ethmoidal foramen | ☐ Groove for middle meningeal artery | ☐ Nasolacrimal canal |
| ☐ Anterior nasal spine | ☐ Groove for sigmoid sinus | ☐ Palatovaginal canal |
| ☐ Carotid canal | ☐ Groove for superior petrosal sinus | ☐ Petrotympanic fissure |
| ☐ Coronoid process | ☐ Hypoglossal (anterior condylar) canal | ☐ Posterior superior alveolar foramen |
| ☐ Depressions (foveae) for arachnoid granulations | ☐ Incisive foramen | ☐ Postglenoid tubercle |
| ☐ Digastric fossa | ☐ Inferior orbital fissure | ☐ Prechiasmatic groove |
| ☐ Digastric notch | ☐ Inferior temporal line | ☐ Pterygoid canal |
| ☐ Dorsum sellae | ☐ Infraorbital foramen | ☐ Pterygoid fovea |
| ☐ External oblique line | ☐ Infraorbital groove | ☐ Pterygoid hamulus |
| ☐ Foramen cecum | ☐ Internal acoustic meatus | ☐ Scaphoid fossa |
| ☐ Foramen ovale | ☐ Internal occipital crest | ☐ Sphenoid spine |
| ☐ Foramen spinosum | ☐ Lambda | ☐ Styloid process |
| ☐ Frontal crest | ☐ Lateral pterygoid plate | ☐ Stylomastoid foramen |
| ☐ Greater palatine foramen | ☐ Lingula | ☐ Sublingual fossa |
| ☐ Groove for auditory tube | ☐ Mental foramen | ☐ Trigeminal depression |
| ☐ Groove for greater petrosal nerve | ☐ Mental spine | ☐ Wormian bones |
| ☐ Groove for inferior petrosal sinus | ☐ Mylohyoid groove | ☐ Zygomaticotemporal foramen |

6. Answer keys

R1. Skull frontal view, orbit, nose

1 nasion · 2 frontonasal suture · 3 frontomaxillary suture · 4 middle nasal concha · 5 frontozygomatic suture · 6 inferior nasal concha · 7 fossa for lacrimal sac · 8 zygomaticomaxillary suture · 9 vomer · 10 frontal bone · 11 fossa for lacrimal gland · 12 supraorbital notch · 13 anterior ethmoidal foramen · 14 lesser wing of sphenoid · 15 posterior ethmoidal foramen · 16 optic canal · 17 superior orbital fissure · 18 greater wing of sphenoid · 19 anterior lacrimal crest · 20 fossa for lacrimal sac · 21 ethmoid bone · 22 inferior orbital fissure · 23 posterior lacrimal crest · 24 zygomaticofacial foramen · 25 infraorbital foramen · 26 zygomatic bone · 27 maxillary bone · 28 nasal bones · 29 anterior nasal aperture · 30 posterior nasal aperture · 31 posterior nasal spine · 32 anterior nasal spine · 33 horizontal plate of palatine bone · 34 palatal process of maxilla

R2. Lateral view, posterior view, vault

1 vertex · 2 superior temporal line · 3 bregma · 4 coronal suture · 5 squamous suture · 6 pterion · 7 inferior temporal line · 8 glabella · 9 lambda · 10 lambdoid suture · 11 nasion · 12 asterion · 13 sphenofrontal suture · 14 inion · 15 occipitomastoid suture · 16 zygomatic arch · 17 tympanic portion of temporal bone · 18 parietomastoid suture · 19 sphenosquamosal suture · 20 mastoid process · 21 sagittal suture · 22 parietal foramina · 23 parietal eminence · 24 wormian bones · 25 highest nuchal line · 26 superior nuchal line · 27 inion · 28 inferior nuchal line · 29 external occipital crest · 30 groove for superior sagittal sinus · 31 depressions for arachnoid granulations

R3. Skull base and cranial fossae

1 transverse palatal suture · 2 incisive foramen · 3 greater palatine foramen · 4 median palatal suture · 5 maxillary tuberosity · 6 lesser palatine foramen · 7 mastoid notch · 8 occipital groove · 9 pterygomaxillary fissure · 10 pterygomaxillary fissure · 11 maxillary tuberosity · 12 infratemporal crest · 13 cribriform plate · 14 anterior cranial fossa · 15 sella turcica · 16 middle cranial fossa · 17 clivus · 18 posterior cranial fossa

R4. Parietal, frontal, occipital (external)

1 groove for superior sagittal sinus · 2 parietal bone · 3 grooves for middle meningeal artery · 4 depressions for arachnoid granulations · 5 squamous portion (frontal) · 6 supraorbital notch/foramen · 7 metopic suture · 8 glabella · 9 superciliary arch · 10 zygomatic process of frontal bone · 11 trochlear fossa · 12 fossa for lacrimal gland · 13 orbital portion (frontal) · 14 nasal portion (frontal) · 15 foramen cecum · 16 frontal crest · 17 foramen magnum · 18 jugular process · 19 occipital condyle · 20 posterior condylar canal · 21 jugular notch · 22 pharyngeal tubercle · 23 external occipital crest · 24 squamous portion (occipital)

R5. Occipital (internal) and temporal

1 jugular foramen · 2 groove for sigmoid sinus · 3 hypoglossal canal · 4 groove for superior petrosal sinus · 5 jugular tubercle · 6 groove for inferior petrosal sinus · 7 foramen magnum · 8 groove for left transverse sinus · 9 groove for right transverse sinus · 10 internal occipital crest · 11 internal occipital protuberance · 12 infratemporal crest · 13 suprameatal triangle · 14 squamous portion (temporal) · 15 zygomatic process (temporal) · 16 petromastoid portion · 17 tympanic portion · 18 mastoid foramen · 19 mastoid process · 20 styloid process · 21 supramastoid crest · 22 occipital groove · 23 digastric notch · 24 mastoid process · 25 lesser petrosal hiatus and groove · 26 groove for sigmoid sinus · 27 greater petrosal hiatus and groove

R6. Temporal: TMJ region, petrous part, foramina

1 postglenoid tubercle · 2 mandibular fossa · 3 external acoustic meatus · 4 articular eminence · 5 squamotympanic fissure · 6 petrosquamous fissure · 7 petrotympanic fissure · 8 trigeminal depression · 9 arcuate eminence · 10 tegmen tympani · 11 foramen ovale · 12 foramen spinosum · 13 foramen lacerum · 14 carotid canal · 15 tympanic canaliculus · 16 stylomastoid foramen · 17 jugular fossa

R7. Sphenoid

1 anterior clinoid process · 2 jugum · 3 lesser wing · 4 superior orbital fissure · 5 prechiasmatic groove · 6 greater wing · 7 orbital surface (sphenoid) · 8 foramen spinosum · 9 carotid groove · 10 spine of sphenoid · 11 pituitary (hypophyseal) fossa · 12 sella turcica · 13 posterior clinoid process · 14 tuberculum sellae · 15 dorsum sellae · 16 optic canal · 17 foramen rotundum · 18 lingula (sphenoid) · 19 inferior orbital fissure · 20 lateral pterygoid plate · 21 foramen ovale · 22 medial pterygoid plate · 23 spine of sphenoid · 24 scaphoid fossa · 25 pterygoid canal · 26 vaginal process · 27 pterygoid hamulus · 28 pterygoid fossa

R8. Pterygoid canals and ethmoid

1 palatovaginal canal · 2 pterygoid hamulus · 3 vomerovaginal canal · 4 crista galli · 5 cribriform plate · 6 ethmoid air cells · 7 superior nasal concha · 8 orbital plate (ethmoid) · 9 middle nasal concha · 10 perpendicular (septal) plate · 11 posterior ethmoidal foramen · 12 anterior ethmoidal foramen

R9. Maxilla

1 frontal process (maxilla) · 2 orbital surface (maxilla) · 3 infraorbital foramen · 4 nasal notch · 5 zygomatic process (maxilla) · 6 anterior nasal spine · 7 incisive fossa · 8 incisive fossa (palatal) · 9 canine fossa/eminence · 10 palatal process · 11 alveolar process · 12 maxillary tuberosity · 13 posterior superior alveolar foramina · 14 anterior lacrimal crest · 15 infraorbital sulcus (groove)

R10. Palatine and zygomatic

1 orbital process (palatine) · 2 vertical plate · 3 posterior nasal spine · 4 pyramidal process · 5 horizontal plate · 6 greater palatine foramen · 7 lesser palatine foramen · 8 frontal process (zygomatic) · 9 orbital surface (zygomatic) · 10 temporal process · 11 maxillary process

R11. Zygomatic foramina and mandible

1 zygomatic foramen · 2 zygomaticofacial foramen · 3 zygomaticotemporal foramen · 4 internal oblique line · 5 mandibular notch · 6 neck of condyle · 7 condylar head · 8 coronoid notch · 9 retromolar fossa · 10 ramus · 11 external oblique line · 12 gonial angle · 13 symphysis menti · 14 body · 15 incisive fossa (mandible) · 16 mental tubercles · 17 mental foramen · 18 coronoid process · 19 mental protuberance · 20 pterygoid fovea · 21 sublingual fossa · 22 lingula of mandible · 23 mandibular foramen · 24 mylohyoid sulcus (groove) · 25 submandibular fossa · 26 lingual foramen · 27 mylohyoid line

R12. Mandible (lingual), lacrimal, nasal

1 digastric fossa · 2 mental spines · 3 lacrimal groove · 4 posterior lacrimal crest · 5 lacrimal hamulus · 6 nasal bone

R13. Conchae and vomer

1 superior nasal concha · 2 inferior nasal concha · 3 middle nasal concha · 4 ala of vomer · 5 vomerovaginal canal

Photo recall

Q1 Mylohyoid groove (mandible) · Q2 Zygomaticotemporal foramen (zygomatic) · Q3 Groove for lesser petrosal nerve (temporal) · Q4 Pterygoid canal (sphenoid) · Q5 Sublingual fossa (mandible) · Q6 Groove for greater petrosal nerve (temporal) · Q7 External oblique line (mandible) · Q8 Pterygoid hamulus (sphenoid) · Q9 Groove for middle meningeal artery (parietal or temporal) · Q10 Lateral pterygoid plate (sphenoid) · Q11 Coronoid process (mandible) · Q12 Lingula (mandible) · Q13 Posterior superior alveolar foramen (maxilla) · Q14 Lambda (parietal, occipital) · Q15 Anterior nasal spine (maxilla) · Q16 Postglenoid tubercle (temporal) · Q17 Foramen ovale (sphenoid) · Q18 Foramen ovale (sphenoid) · Q19 Frontal crest (frontal) · Q20 Infraorbital groove (maxilla) · Q21 Groove for superior petrosal sinus (temporal) · Q22 Nasolacrimal canal (lacrimal, maxilla) · Q23 Carotid canal (temporal) · Q24 Hypoglossal (anterior condylar) canal (occipital) · Q25 Mental spine (mandible) · Q26 Greater palatine foramen (palatine) · Q27 Mylohyoid line (mandible) · Q28 Wormian bones (varies (parietal/occipital/temporal)) · Q29 Incisive foramen (maxilla) · Q30 Groove for inferior petrosal sinus (temporal) · Q31 Sphenoid spine (sphenoid) · Q32 Groove for sigmoid sinus (temporal) · Q33 Anterior clinoid process (sphenoid) · Q34 Groove for auditory tube (temporal) · Q35 Internal acoustic meatus (temporal) · Q36 Styloid process (temporal) · Q37 Palatovaginal canal (palatine, sphenoid) · Q38 Anterior ethmoidal foramen (ethmoid, frontal) · Q39 Foramen spinosum (sphenoid) · Q40 Stylomastoid foramen (temporal) · Q41 Hypoglossal (anterior condylar) canal (occipital) · Q42 Depressions (foveae) for arachnoid granulations (parietal/frontal (varies)) · Q43 Foramen cecum (frontal) · Q44 Zygomaticotemporal foramen (zygomatic) · Q45 Petrotympenic fissure (temporal) · Q46 Inferior orbital fissure (sphenoid, maxilla) · Q47 Digastric fossa (mandible) · Q48 Prechiasmatic groove (sphenoid) · Q49 Mental foramen (mandible) · Q50 Inferior temporal line (parietal) · Q51 Infraorbital foramen (maxilla) · Q52 Foramen spinosum (sphenoid) · Q53 Internal occipital crest (occipital) · Q54 Trigeminal depression (temporal) · Q55 Pterygoid fovea (mandible) · Q56 Digastric notch (temporal) · Q57 Dorsum sellae (sphenoid) · Q58 Scaphoid fossa (sphenoid)